

DAB60 power assembly

Rev B0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | DAB-D60 60 kW 100 kHz isolated DAB: 2 x CBB011M12GM4T, 11:12 transformer, two DC ports

01 Bridge 1 power stage

CBB011M12GM4T, port 1 (HV1), 3 x C4AQUEW5450A3BJ DC link, DD600N16K reverse clamp, 12 RC snubbers + 10 ceramics, AC out

File: sheets/01_b1_power.kicad_sch

02 Bridge 1 gate drive leg A

UCC21710 + UCC14241-Q1, +15/-4 V, Ron/Roff 1/0.27 ohm, DESAT, interlock H1<->L1; module NTC on the L1 driver AIN/APWM

File: sheets/02_b1_gatea.kicad_sch

03 Bridge 1 gate drive leg B

UCC21710 + UCC14241-Q1, +15/-4 V, Ron/Roff 1/0.27 ohm, DESAT, interlock H2<->L2

File: sheets/03_b1_gateb.kicad_sch

04 Bridge 1 CELL interface + logic

CELL C1: PWM1-4 / EN RS-422 in (fail-safe), FLT, RDY, ID 20.0k trip latch, PWM gating, +24V UV, 5 V / 3.3 V from the CELL +24V

File: sheets/04_b1_if.kicad_sch

05 Bridge 1 sensing

HOB 130-P primary current, HV1 side -> AN1 (AMC3330), OC +/-210 A (ISO7720F), |Ldc| 5 A on AN1, AN2/AN3 temperatures

File: sheets/05_b1_sense.kicad_sch

06 Bridge 2 power stage

CBB011M12GM4T, port 2 (HV2), 3 x C4AQQEW5650A3BJ DC link, DD600N16K reverse clamp, 12 RC snubbers + 10 ceramics, AC out

File: sheets/06_b2_power.kicad_sch

07 Bridge 2 gate drive leg A

UCC21710 + UCC14241-Q1, +15/-4 V, Ron/Roff 1/0.27 ohm, DESAT, interlock H1<->L1; module NTC on the L1 driver AIN/APWM

File: sheets/07_b2_gatea.kicad_sch

08 Bridge 2 gate drive leg B

UCC21710 + UCC14241-Q1, +15/-4 V, Ron/Roff 1/0.27 ohm, DESAT, interlock H2<->L2

File: sheets/08_b2_gateb.kicad_sch

09 Bridge 2 CELL interface + logic

CELL C2: PWM1-4 / EN RS-422 in (fail-safe), FLT, RDY, ID 40.2k trip latch, PWM gating, +24V UV, 5 V / 3.3 V from the CELL +24V

File: sheets/09_b2_if.kicad_sch

10 Bridge 2 sensing

HOB 130-P secondary current, HV2 side -> AN1 (AMC3330), OC +/-210 A (ISO7720F), |Ldc| 5 A on AN1, AN2/AN3 temperatures

File: sheets/10_b2_sense.kicad_sch

11 Transformer, inductor, DC-block link

Transformer 11:12 and series inductor (CUSTOM, cold plate), bolted link = DC-blocking capacitor position (bank not fitted)

File: sheets/11_magnetics.kicad_sch

12 Port A power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/12_portA_power.kicad_sch

13 Port A sensing and hold logic

VA bus side, VAX terminal side (AMC3330), IA current (AMC3302), two-channel hold logic, proof-test injection, readback RB1/RB2

File: sheets/13_portA_sense.kicad_sch

14 Port B power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/14_portB_power.kicad_sch

15 Port B sensing and hold logic

VB bus side, VBX terminal side (AMC3330), IB current (AMC3302), two-channel hold logic, proof-test injection, readback RB1/RB2

File: sheets/15_portB_sense.kicad_sch

16 Connectors, port OV trip, coolant loop

PORT 26-way + ID readback, 14-way SYS-IO-AUX, +3V3S, AUX-HV tap, port OV comparators -> OV_OK, coolant loop -> SYS-IO-AUX J801

File: sheets/16_io.kicad_sch

CALCULATED, not bench-validated. Values from sim/out/dab_design/dab_spec.json, checked by design_check() (outputs/DAB60_design_check.txt). Bridge 1 -> CTRL-C2000 C1, bridge 2 -> C2 (HRPWM). Ports via port.port(): port 1 = tag A (HV1), port 2 = tag B (HV2).

Fail-safe off: received pairs biased 470R/120R/470R -> undriven VID <= -0.29 V (receiver -0.2 V) = LOW = off; no 3.3 V -> UCC14241 ENA low, PWM/EN pulled low; +24V_GD below 19.7 V -> EN_DRV and PWM forced low before the bias UVLO.

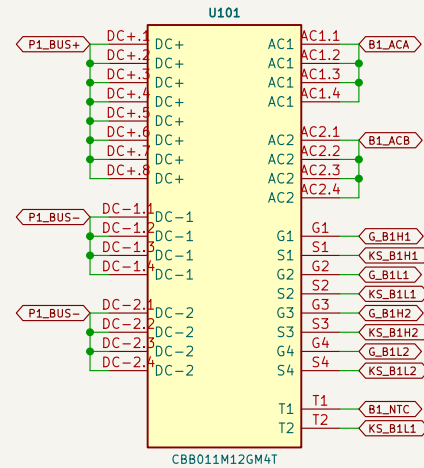
Local trips, latched until EN low (no firmware): transformer OC +211/-210 A (0.78 us), |Ldc| 5.0 A (tau 1 ms), port OV 986 / 937 V (37 us), +24V UV, driver UVLO; DESAT gates all PWMs of its bridge. FLT reports all, also with EN low; RDY is driven.

AN1 = 5.005 mV/A on 1.44 V CM (AMC3330, 0.2 ohm). Domains: PELV, HV1, HV2, PE + 8 gate drivers. Isolators: UCC21710, UCC14241-Q1, UCC12050, AMC3330, ISO7720F, transformer, port AMC/TPSI, contactors.

ECO-10 (layout): HV1/HV2-PELV reinforced >= 1000 V DC at every isolator footprint (8 mm parts: PD1 coating or slots); the HOB 130-P sits inside HV1/HV2 (functional). HV1-HV2 via the transformer, 1850 V DC basis.

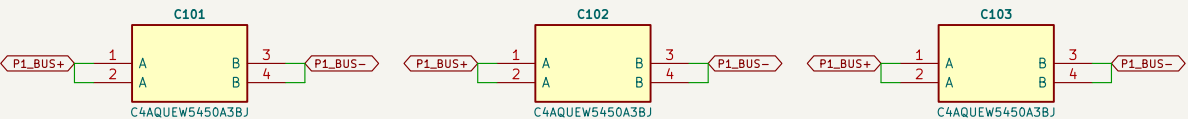
Not bench-validated. Domains PELV, HV1, HV2, PE, GD_B<sw> (8 gate drivers). Sheet 1 Schematics authored by Chinmoy Bhuyan Interfaces: CELL x2 (C1, C2), PORT, SYS-IO-AUX harness + coolant loop (gen/interfaces.py)		
Sheet: / File: DAB60.kicad_sch		
Title: DAB60 power assembly – Cover / Hierarchy		
Size: A3	Date: 2026-10-04	Rev: B0
KiCad E.D.A. 10.0.6		Id: 1/17

CBB011M12GM4T FULL-BRIDGE MODULE (PORT 1 SIDE)



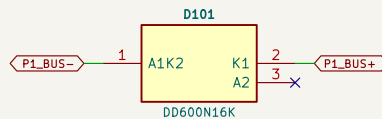
Switch 1/2 = leg A (AC1), 3/4 = leg B (AC2). Kelvin sources S1–S4 go to the drivers only (COM).
NTC T1/T2 is read by the switch–2 driver (AIN referenced to S2), as on the CRD. Drains of 1/3 = DC+.

DC LINK 3 X C4AQEW5450A3BJ = 135 UF



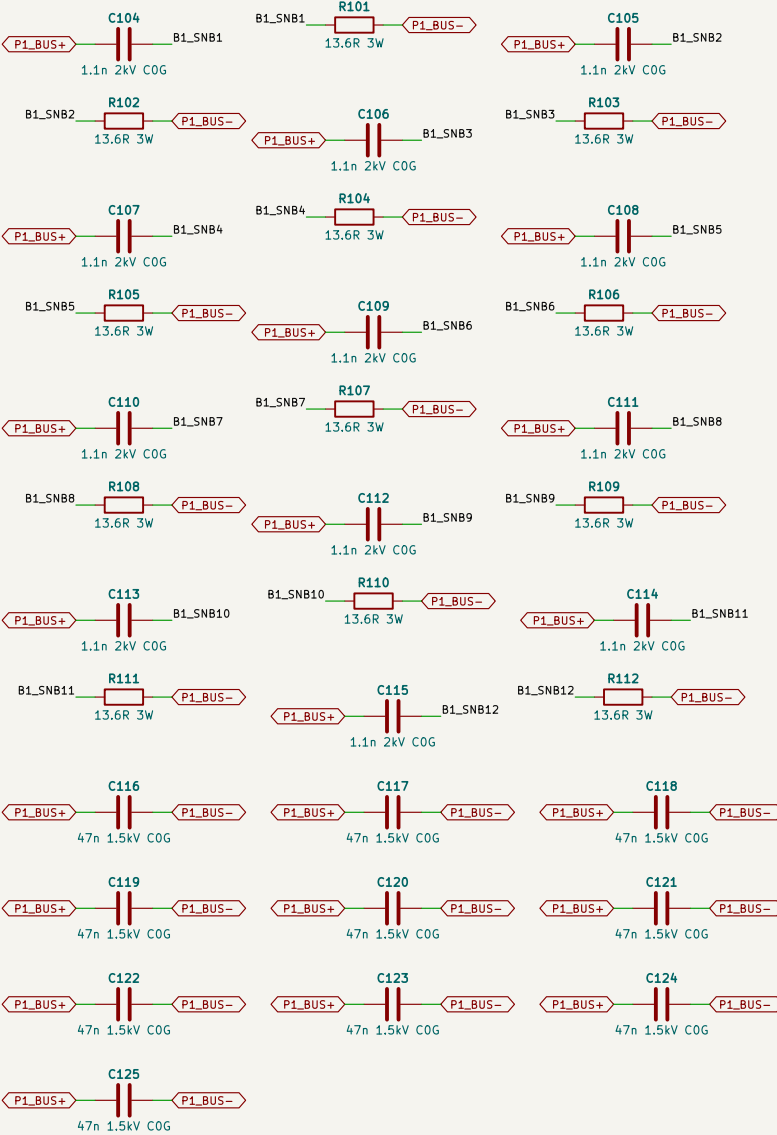
Bank ripple 73 A rms, 24.4 A per capacitor (rating 34 A); hot spot 73 C → 1261 V allowed vs 950 V.
Ratings vs stresses: design_check(J).

DC-LINK REVERSE CLAMP (DD600N16K, ONE ARM)



Cathode (2) to DC+, anode (1) to DC– at the bank terminals; terminal 3 unused. Reverse-biased in operation.
Port short: carries 15.7 kA / 16.5 kA2s after the bank reverses (I2t 1.8 MA2s) instead of the module body diodes. CHASSIS-MOUNTED, insulated base plate (basic, to PE frame).

HF RC SNUBBERS 12 X (13.6 OHM 3 W + 1.1 NF) AND 10 X 47 NF TERMINAL CERAMICS



Across DC+/DC– at the module pins (CRD UG Fig. 22 / 19 networks). Spread over both legs in layout:
resistors: pulse-rated wide-terminal 2512 with a thermal path to the plane (dab.spec. 1.21 W worst).

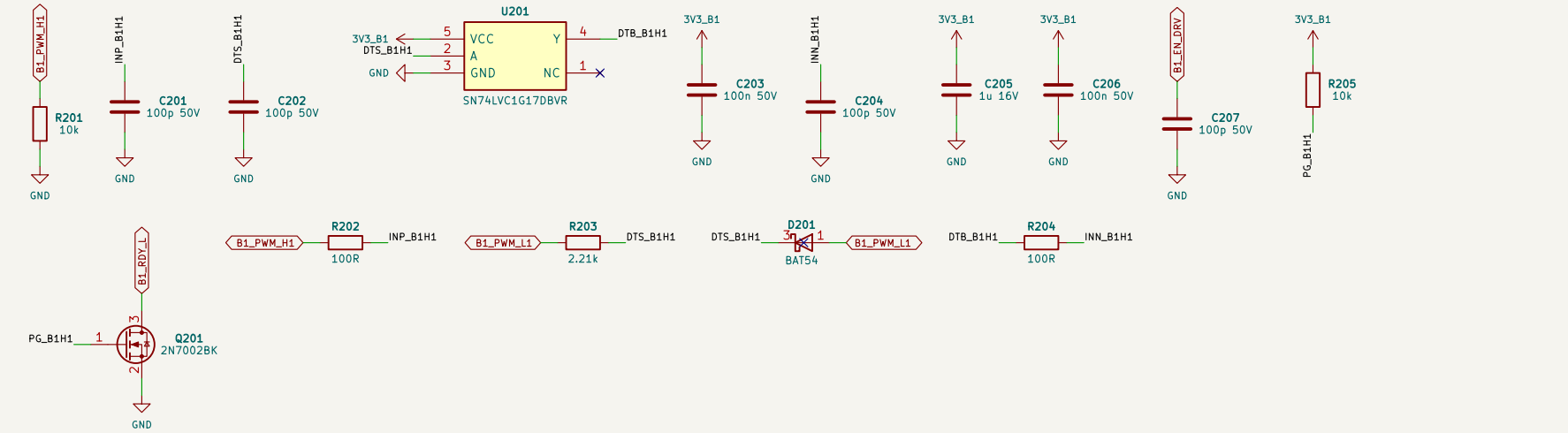
Not bench-validated. Domains PELV, HV1, HV2, PE, GD_B<sw> (8 gate drivers).
Sheet 2
Schematics authored by Chinmoy Bhuyan
Interfaces: CELL x2 (C1, C2), PORT, SYS–IO–AUX harness + coolant loop (gen/interfaces.py)

Sheet: /01 Bridge 1 power stage/
File: 01_b1_power.kicad_sch
Title: DAB60 power assembly – Bridge 1 power stage
Size: A2 Date: 2026–10–04 Rev: B0
KiCad E.D.A. 10.0.6 Id: 2/17

MILLER MARGIN (GDRV REV 4 MODEL, ROFF 0.27 OHM, CLMPI DIRECT)

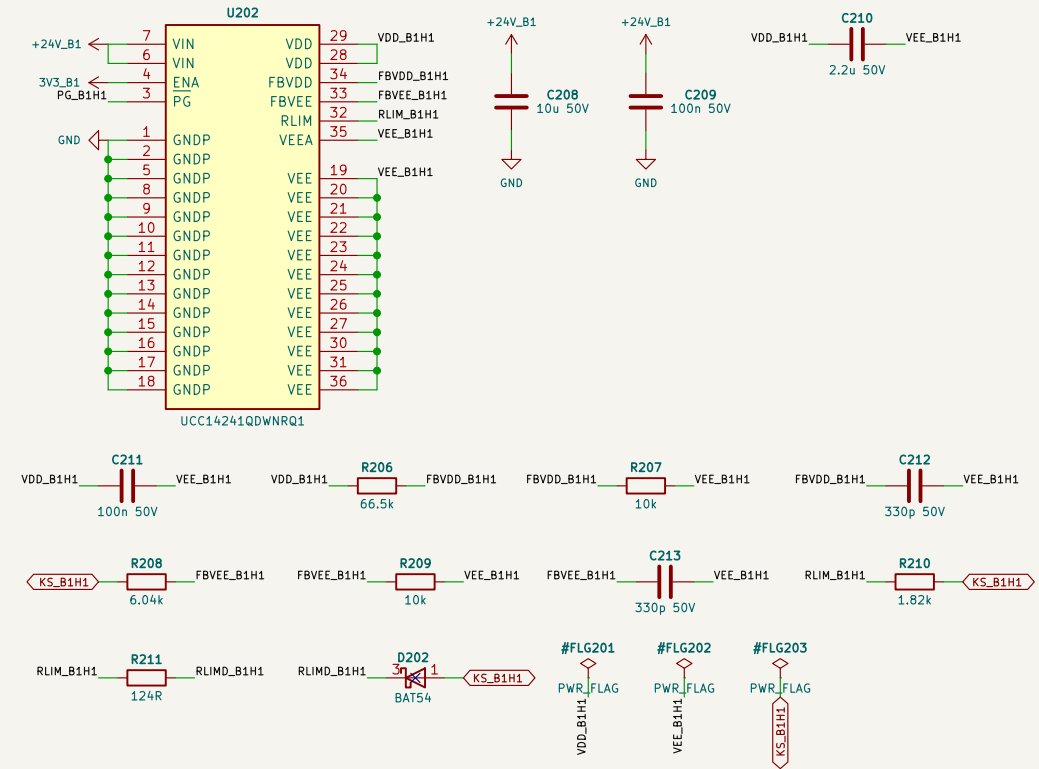
Die VGS of the OFF switch while its partner turns on: -0.55 V at 56.7 V/ns (dab_spec, I_M 2.0 A), +0.87 V at 80 V/ns (module max, 2.9 A) vs VGS(th) 1.44 V at 175 °C: fine at the DAB rate, marginal at 80 V/ns (gdrv's own case, R_{off} 1 ohm: +1.19 V). Crss(800 V) understates the low-voltage Cgd: confirm by DPT.

B1H1: LOGIC SIDE



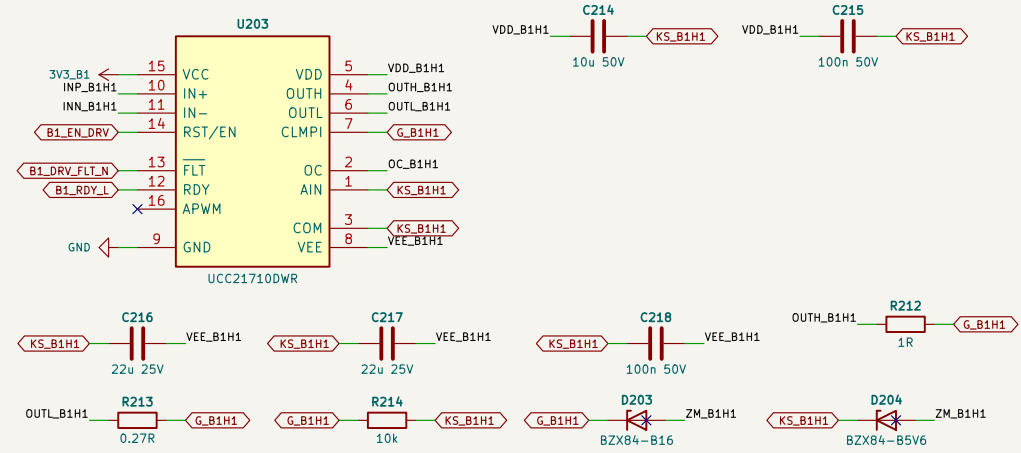
IN+ = own PWM; IN- = complementary switch, falling edge stretched 2.21k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

B1H1: ISOLATED BIAS +15 V / -4 V



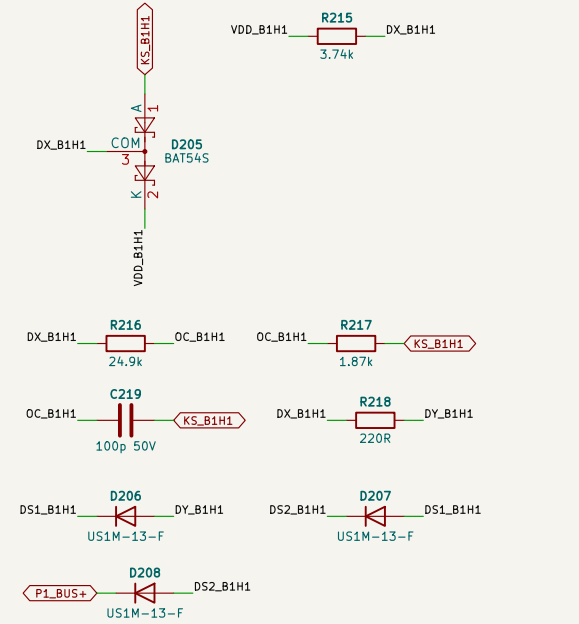
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

B1H1: DRIVER AND GATE



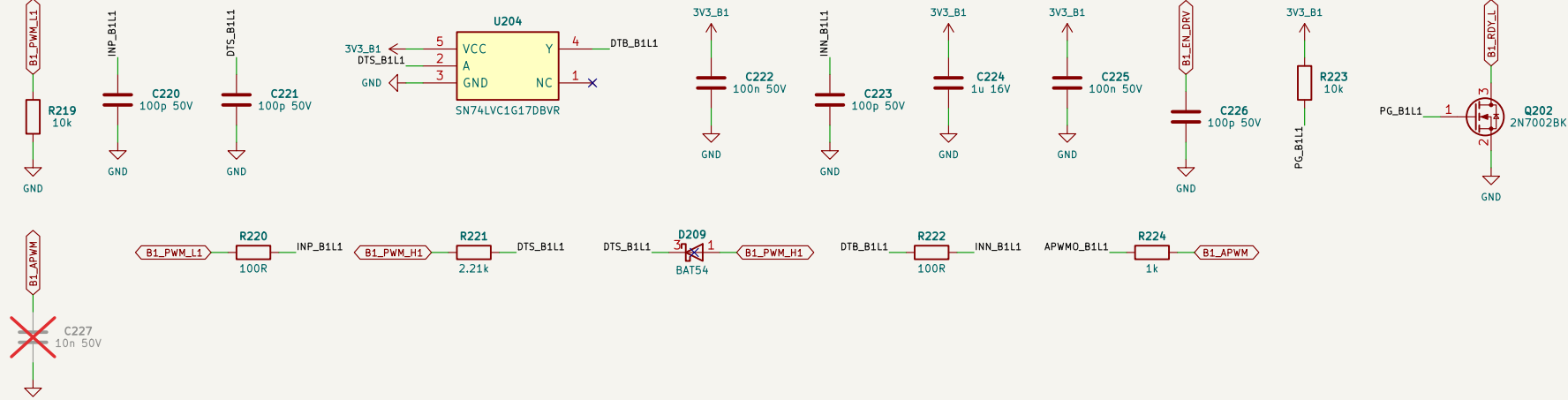
CLMPI direct to the gate (active Miller clamp). Kelvin source = COM.
Zener pair clamps the gate to about +17 V / -6.6 V when the driver cannot.

B1H1: DESAT ON OC



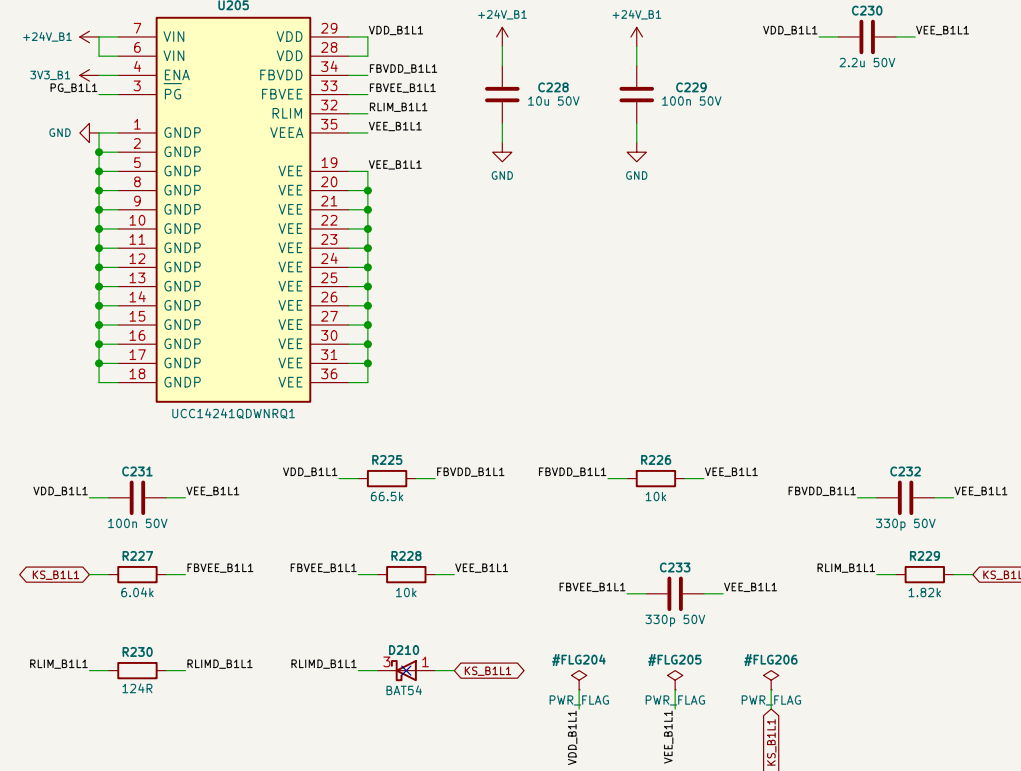
VDD-R1-DX; DX-R2-0C-R3/Cb1k-COM; DX-Rs-3 x US1M-drain (SLUSD43B Fig 9-8). BAT54S keeps DX between COM and VDD against dv/dt through the string.

B1L1: LOGIC SIDE



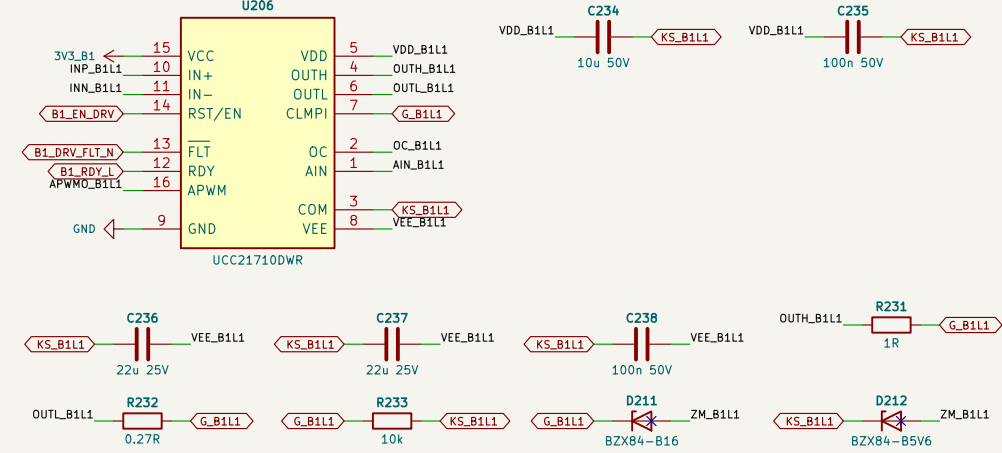
IN+ = own PWM; IN- = complementary switch, falling edge stretched 2.21k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

B1L1: ISOLATED BIAS +15 V / -4 V



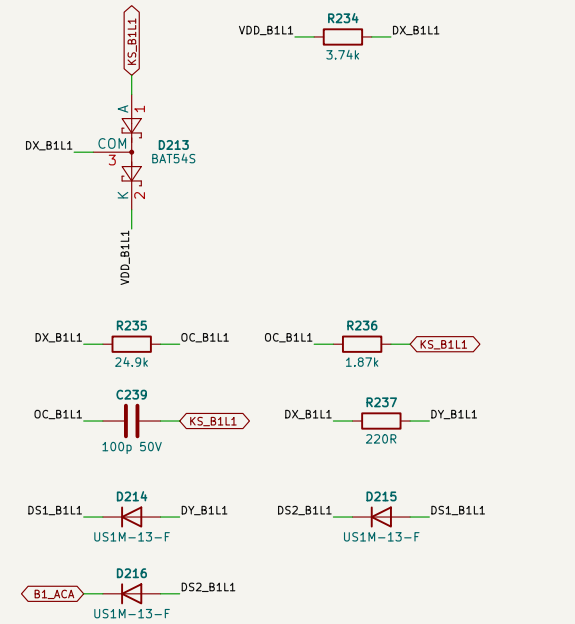
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
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B1L1: DRIVER AND GATE



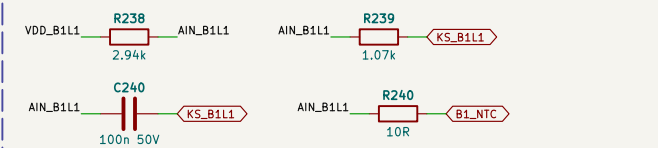
CLMPI direct to the gate (active Miller clamp). Kelvin source = COM.
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B1L1: DESAT ON OC



VDD-R1-DX; DX-R2-OC-R3/Cb1k-COM; DX-Rs-3 x US1M-drain (SLUSD43B Fig 9-8)
BAT54S keeps DX between COM and VDD against dv/dt through the string.

B1L1: NTC $\rightarrow$ AIN



VDD=2.94k-AIN=1.07k-COM biases the NTC ~10x harder than the 203 uA source alone; AIN stays inside 0.6-4.5 V (design_check), -0.5 %/C APWM duty at 100 C.

Not bench-validated. Domains PELV, HV1, HV2, PE, GD_B<sw> (8 gate drivers).
Sheet 3
Schematics authored by Chinmoy Bhuyan
Interfaces: CELL x2 (C1, C2), PORT, SYS-I/O-AUX harness + coolant loop (gen/interfaces.py)

Sheet: /02 Bridge 1 gate drive leg A/
File: 02_b1_gatea.kicad_sch

Title: DAB60 power assembly – Bridge 1 gate drive leg A

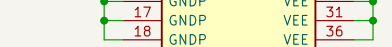
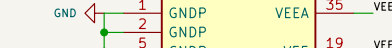
Size: A2	Date: 2026-10-04	Rev: B0
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KiCad E.D.A. 10.0.6 Id: 3/17

Die VGS of the OFF switch while its partner turns on: -0.55 V at 56.7 V/ns (dab_spec, LM 2.0 A), +0.87 V at 80 V/ns (module max, 2.9 A) vs VGS(th) 1.44 V at 175 °C: fine at the DAB rate, marginal at 80 V/ns (gdrv's own case, Roff 1 ohm; +1.19 V). Crss(800 V) understates the low-voltage Cgd: confirm by DPT.



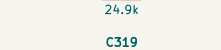
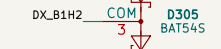
3V3_B1	4	VIN	VDD	34	FBV
PG_B1H2	3	ENA	FBVDD	33	FBV
		PG	FBVEE	32	



3V3_B1	19	VCC
INP_B1H2	10	IN+
INN_B1H2	11	IN-



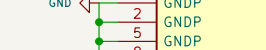
AI



C32



3V3_B1 ← 4 ENA F
PG_B1L2 3 PG F



INF_B1L2	10	IN+
INN_B1L2	11	IN-
B1_EN_DRV	14	RST/EN



The diagram shows the pinout for the 61204021621 component, which is a 40-pin device. The pins are arranged in two rows of 20 pins each. The signals are as follows:

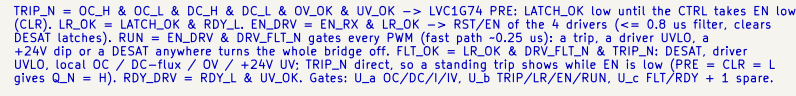
- Pin 1:** +24V_B1
- Pin 2:** GND
- Pin 3:** B1_PWM1_P
- Pin 4:** B1_PWM1_N
- Pin 5:** B1_PWM2_P
- Pin 6:** B1_PWM2_N
- Pin 7:** B1_PWM3_P
- Pin 8:** B1_PWM3_N
- Pin 9:** B1_PWM4_P
- Pin 10:** B1_PWM4_N
- Pin 11:** GND
- Pin 12:** GND
- Pin 13:** GND
- Pin 14:** GND
- Pin 15:** GND
- Pin 16:** GND
- Pin 17:** GND
- Pin 18:** GND
- Pin 19:** GND
- Pin 20:** GND
- Pin 21:** GND
- Pin 22:** GND
- Pin 23:** GND
- Pin 24:** GND
- Pin 25:** B1_FLT_P
- Pin 26:** B1_FLT_N
- Pin 27:** B1_EN_P
- Pin 28:** B1_EN_N
- Pin 29:** B1_RDY
- Pin 30:** B1_ID
- Pin 31:** B1_AN1_P
- Pin 32:** B1_AN1_N
- Pin 33:** B1_AN2_P
- Pin 34:** B1_AN2_N
- Pin 35:** B1_AN3_P
- Pin 36:** B1_AN3_N
- Pin 37:** AGND
- Pin 38:** AGND
- Pin 39:** AGND
- Pin 40:** AGND

The diagram also shows the following components and connections:

- J401:** A 40-pin connector on the left side of the component.
- R401:** A 20.0k resistor connected between pin 1 and pin 2.
- #FLCA01:** A component connected to pin 3.
- PWR_FLAG:** A signal connected to pin 4.
- +24V_B1:** A power supply connected to pin 5.
- AGND:** Ground connections for pins 31, 32, 33, 34, 35, 36, 37, 38, 39, and 40.

The component number 61204021621 is printed at the bottom of the diagram.

TRIP LATCH, ENABLES, FLT (NO FIRMWARE IN THE PATH; INT-17, INT-18)



Each pair: P 470R to GND. N 470R to 3V3. 120R across (= termination). AM26LV32E: VID <= -0.2 V -> output LOW.
 Cable unplugged / pair open: VID <= -0.303 V worst case (3.195 V, 1.5%, r/k k) -> LOW = gate off.
 CTRL unpowered: Its AM26LV32E outputs are high-Z (10(0)F) <= 100 uA -> VID <= -0.293 V -> LOW = off.
 This boards 3.3 V absent: receivers, gates and UCC2210 VCC dead. UCC42241 EMA low -> no gate bias.
 PWM/EN nets held low by 10 k. By the whole CELL cable out this board has no +24V at all.

Each pair: P 470R to GND, N 470R to 3.3 V, 120R across (= termination), AM26LV32E: VID <= -0.2 V -> output LOW.
Cable unplugged / pair open: VID <= -0.303 V worst case (3.135 V, 1 %, ri 4 k) -> LOW = gate off.
CTRL unpowered: its AM26LV31E outputs are high-Z (IO(OFF) <= 100 uA) -> VID <= -0.293 V -> LOW = off.
This board's 3.3 V absent: receivers, gates and UCC21710 VCC dead, UCC14241 ENA low -> no gate bias;
PWM/EN nets held low by 10 k. With the whole CELL cable open this board has no +24V at all.

[illegible]

FLT_OK high = healthy -> FLI_P high (CTRL receiver wired swapped: high = fault); unpowered = high-Z = FAULT at the CTRL's bias. RDY_L = wired-AND of the 4 UCC21710 RDY (VCC/VDD UVLO, 4.99k). CELL RDY is DRIVEN high by an LVOC8 gate (RDY_L & UV_OK, 100R) - the CTRL pulls it down with 100k and tri-states its PWM/EN drivers while RDY is low (INT=15); no 3.3 V here = output low = not ready.

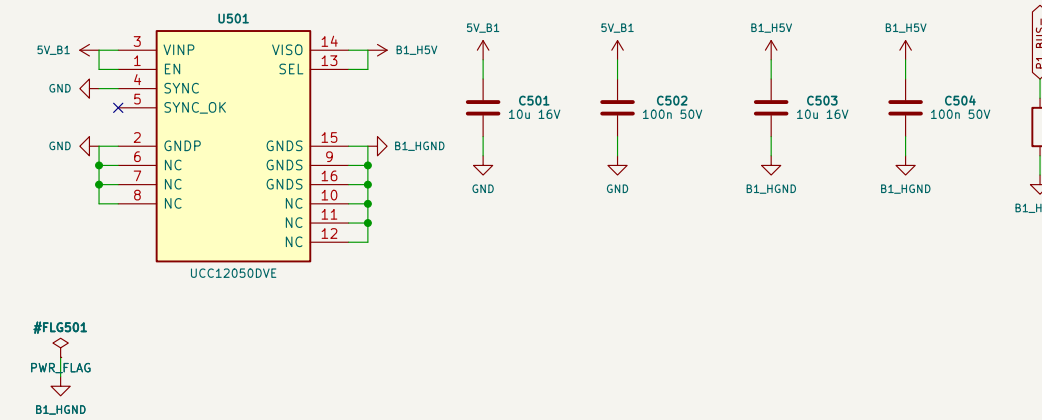
Figure 10 shows the schematic diagram of the TPSE3710DDCR circuit. The chip is represented by a yellow box with pins 1 through 6. Pin 1 is labeled B1_UV_OK. Pin 2 is labeled VDD. Pin 3 is labeled SENSE. Pin 4 is labeled OUT. Pin 5 is labeled GND. Pin 6 is labeled GND. The chip is connected to a +24V_B1 supply. A resistor R421 (100k) is connected between +24V_B1 and pin 2. A resistor R422 (2k) is connected between +24V_B1 and pin 3. A capacitor C411 (1n 50V) is connected between +24V_B1 and pin 4. A capacitor C412 (100n 50V) is connected between +24V_B1 and pin 5. The output OUT is connected to B1_UV_OK.

Trips at 19.70–20.44 V falling (20.16–20.64 V rising, 100k/2.00k 0.1 %): above the UCC14241 UVLO (17.1–18.9 V falling) and below the 21.6 V system minimum, so a vanishing +24V_GD turns every gate off actively (PWM gating + EN, latched) while the gate bias is still regulated, tpd(HL) 18 us.

[illegible]

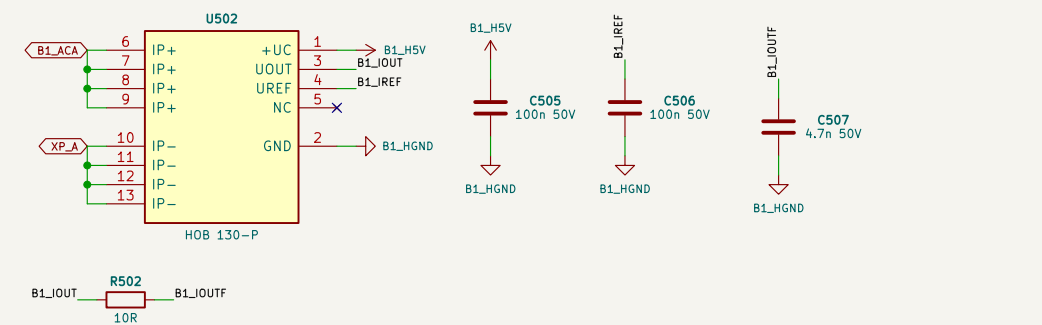
SNVSB49D table 10-3 (5 V, 400 kHz): 10 uH, 2 x 22 uF, 100k / 24.9k -> 5.02 V. EN tied to VIN.
5 V: UCC12050 (HV-side sensor), AMC3330, PELV op amps / comparator / reference; 3.3 V: receivers,
logic, ISO7720F side 2, UCC21710 VCC, bias ENA.

INSULATION: THE HOB 130-P IS NOT THE PELV BARRIER (BASIC 1000 V ONLY)



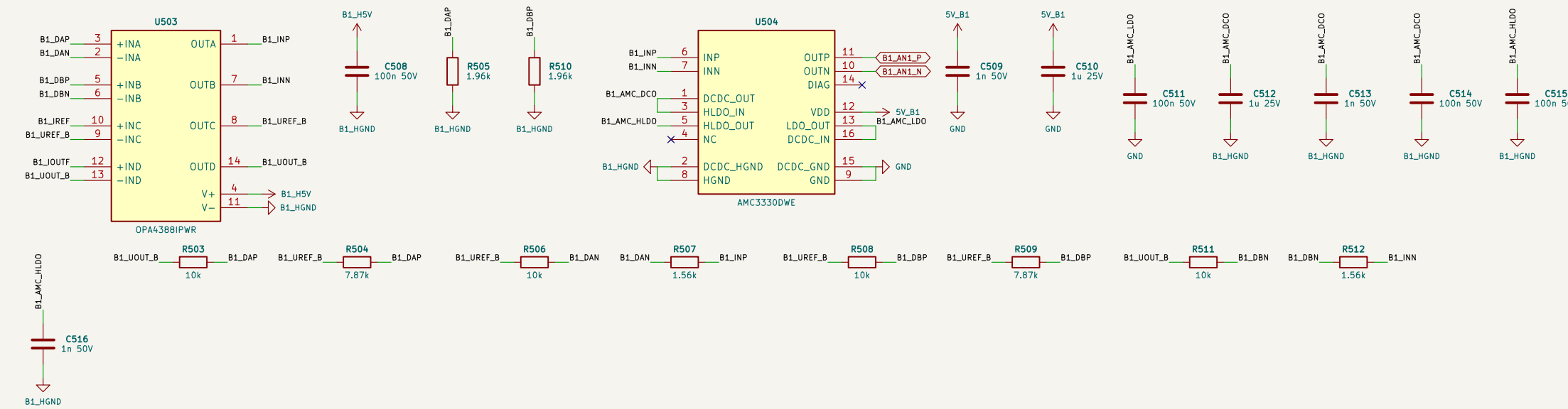
Its secondary lives on the HV1 side, referenced to the bridge DC- (P1_BUS-, one 0R Kelvin tie at the bank), so primary-secondary sees only functional stress. HV1 -> PELV crosses three reinforced parts: UCC12050 power (VIOWM 1697 VDC), AMC3330 AN1 (1700 VDC), ISO7720F OC trips (2121 VDC). ECO-10: reinforced creepage at these footprints (8 mm packages < 10 mm at PD2 / 1000 V: PD1 coating or slots, as PV-PORT).

TRANSFORMER PRIMARY CURRENT: HOB 130-P (DC-1 MHZ, +/-250 A, 8 MV/A)



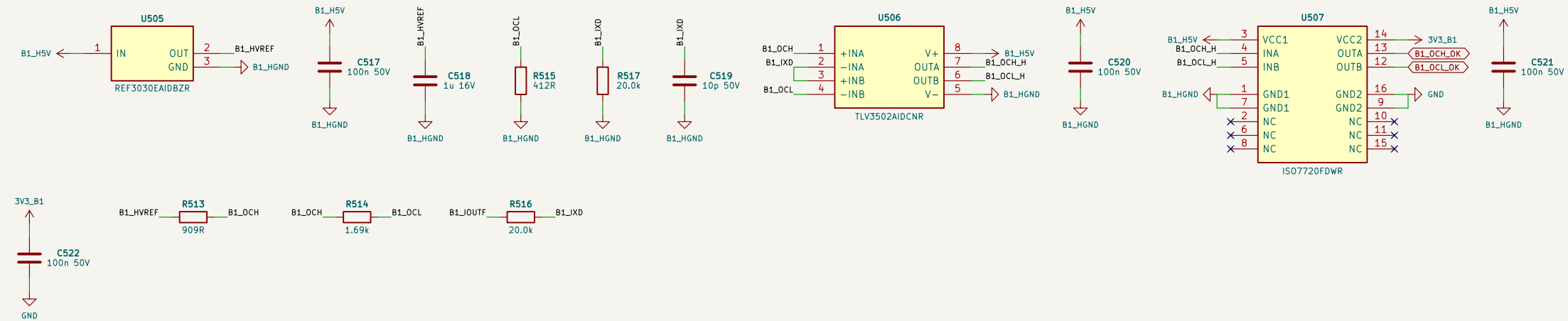
Primary jumper in the AC path B1_ACA -> XP_A (positive = out of leg A). LEM p12 filter 10R / 4.7n on Uout. OPA4388 D buffers Uout (Rout 30 ohm), C buffers Uref (130-300 ohm).

AN1 DRIVER -> AMC3330 (+/-1 V IN, GAIN 2, 1.44 V CM OUT, 0.2 OHM)



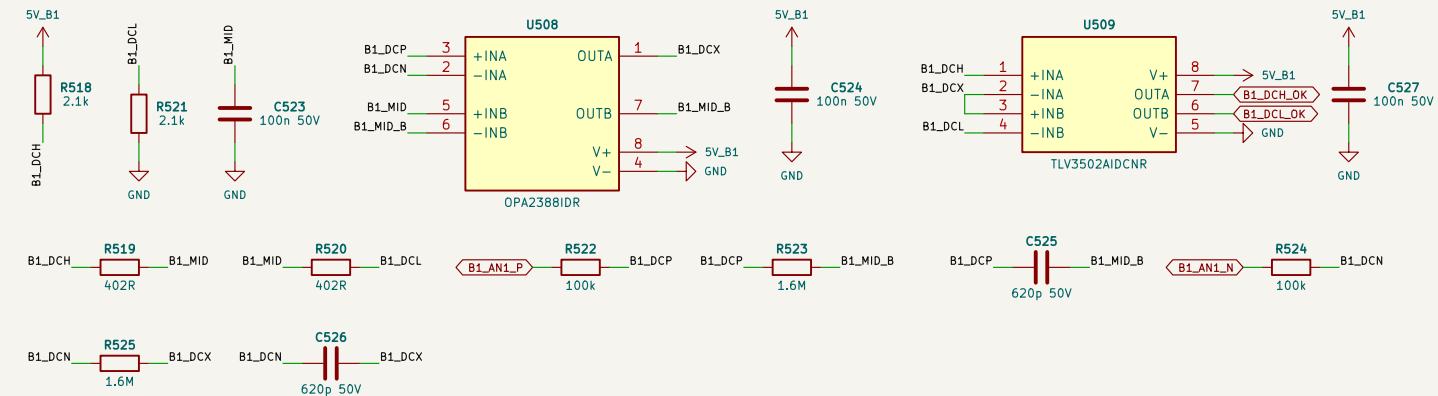
OPA4388 A: $I_{NP} = V_{cm} + k(U_{out} - U_{ref})$, B: $I_{NN} = V_{cm} - k(U_{out} - U_{ref})$; $R_1 = R_3 = 10k$, $R_2 = 1.56k$, + leg divider 7.87k / 1.96k from Uref_B ($V_{cm} 0.500V$, $R_{th} - R_2$). $V_{IN} = 0.3128(U_{out} - U_{ref}) \rightarrow A_{N1} = 5.005$ mV/A differential; 0 A = 0 V, +/-1.25 V at +/-250 A, +/-1.56 V at the HOB rails; CTRL G 0.402 $\rightarrow$ 0.62-1.88 V at its ADC.

OVER-CURRENT WINDOW +/-210 A, LOCAL AND FAST (HV SIDE -> ISO7720F -> TRIP_N)



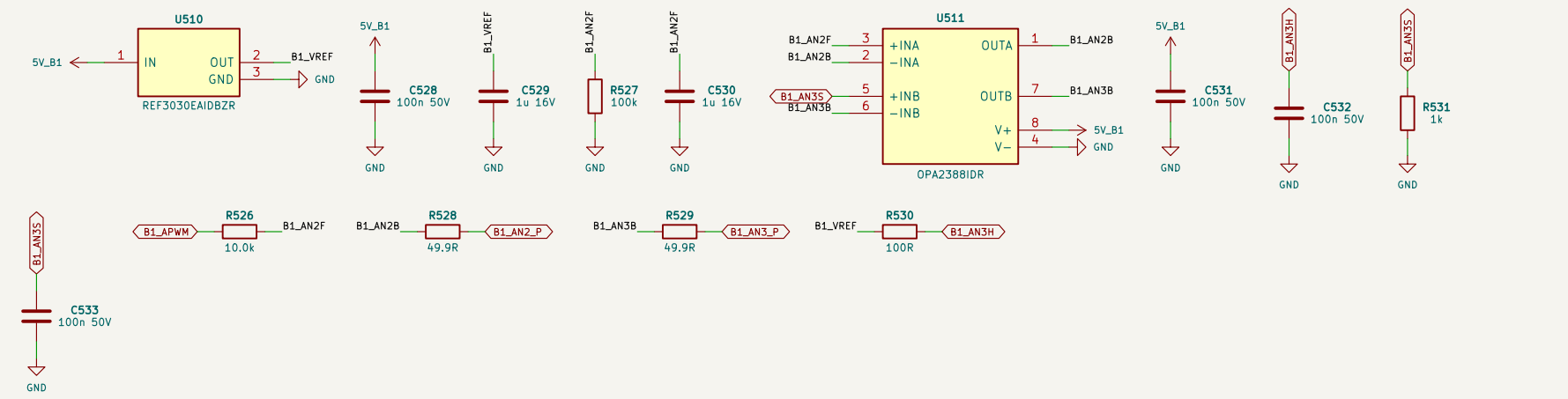
REF3030E ladder 909R/1.69k/412R (0.1 %) on H5V; trips at +211.1 / -209.9 A on Uout/2 (20.0k/20.0k + 10 pF). TLV3502 out HIGH = OK -> ISO7720F (default LOW: a dead HV side reads TRIP) -> OCH_OK / OCL_OK. -0.78 us to the gate.

DC-FLUX WINDOW ± 5 A ON AN1 (PELV, TAU 0.99 MS)



OPA2388 A: MID + 16 x LPF(AN1_P - AN1_N) (100k / 1.60M || 620p COG); B buffers MID, 5V_B ladder 2.1k/402R/402R/2.1k (0.1 % , ratiometric); trips at +5.02 / -5.02 A, TLV3502 out LOW = trip -> TRIP_N, AMC3330 fail-safe (-2.57 V, HV side dead) and an open AN1 line trip it too.

AN2 MODULE TEMPERATURE, AN3 TRANSFORMER WINDING HOT SPOT (PELV)



AN2: APWM (L1 driver, 400 kHz, duty 88-10 % for AIN 0.6-4.5 V) -> 10.0k / 100k / 1 uF (tau 10 ms) -> follower: 0.9 x 3.3 V x duty, 0.2-2.8 V, AN3: REF3030E 3.0 V -> 100R -> NTC -> 1k -> GND, follower, 0-2.73 V. Outputs 49.9R (CTRL: <= 100 R source), single-ended on AGND.

Not bench-validated. Domains PELV, HV1, HV2, PE, GD_8<sw> (8 gate drivers).
Sheet 6
Schematics authored by Chinmoy Bhuyan
Interfaces: CELL x2 (C1, C2), PORT, SYS-I/O-AUX harness + coolant loop (gen/interfaces.py

Sheet: /05 Bridge 1 sensing/
File: 05_b1_sense.kicad_sch

Title: DAB60 power assembly – Bridge 1 sensing

Size: A2 Date: 2026-10-04

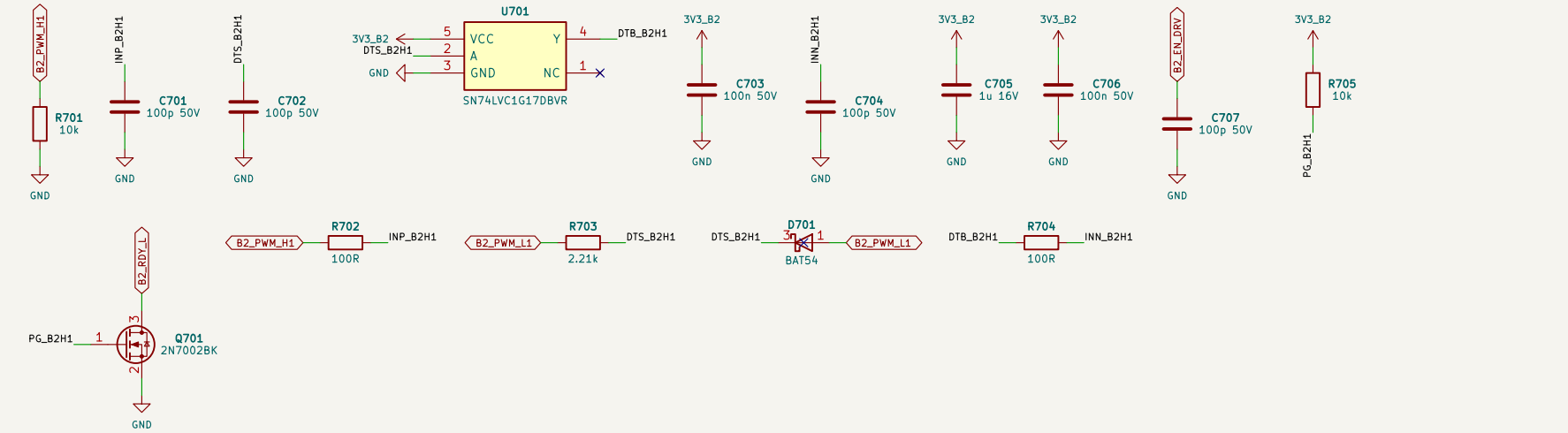
Rev: B0

Id: 6/17

MILLER MARGIN (GDRV REV 4 MODEL, ROFF 0.27 OHM, CLMPI DIRECT)

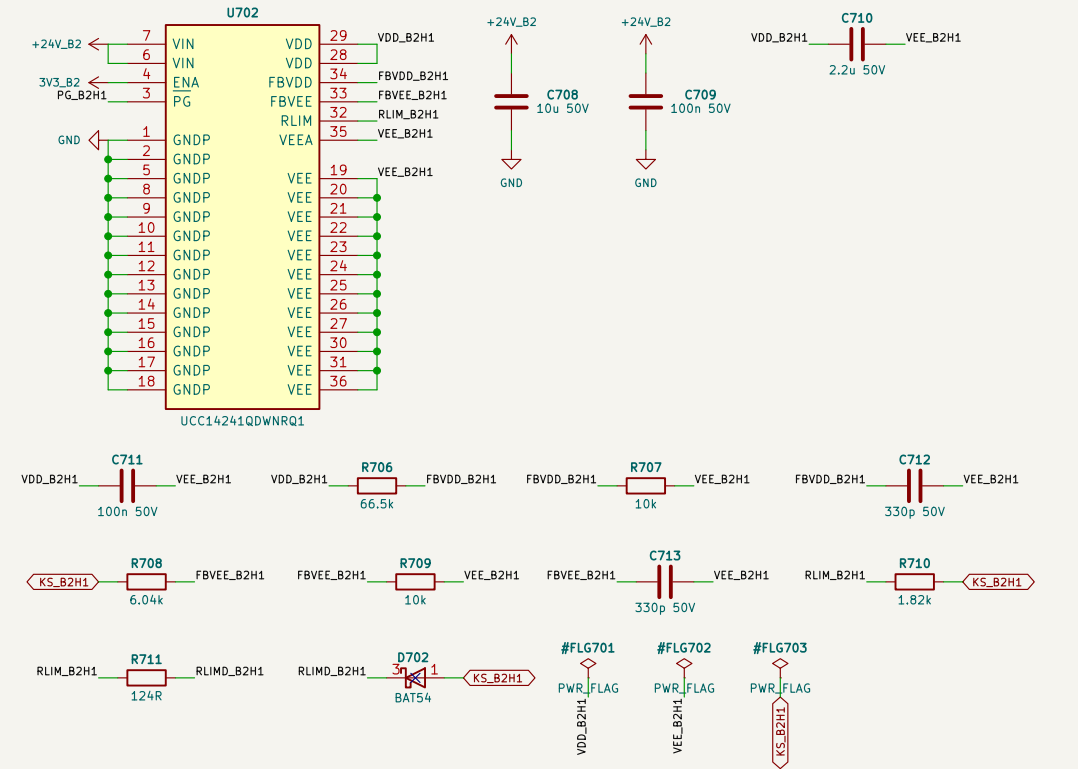
Die VGS of the OFF switch while its partner turns on: -0.55 V at 56.7 V/ns (dab_spec, $I_M\ 2.0\text{ A}$), $+0.87\text{ V}$ at 80 V/ns (module max, 2.9 A) vs VGS(th) 1.44 V at 175 C : fine at the DAB rate, marginal at 80 V/ns (gdrv's own case, $R_{off}\ 1\text{ ohm}$: $+1.19\text{ V}$). Crss(800 V) understates the low-voltage Cgd: confirm by DPT.

B2H1: LOGIC SIDE



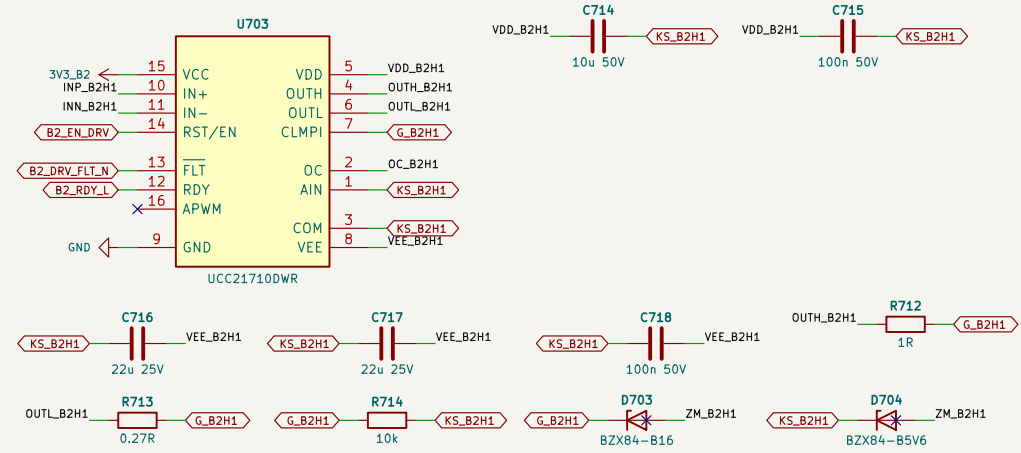
IN+ = own PWM; IN- = complementary switch, falling edge stretched 2.21k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

B2H1: ISOLATED BIAS +15 V / -4 V



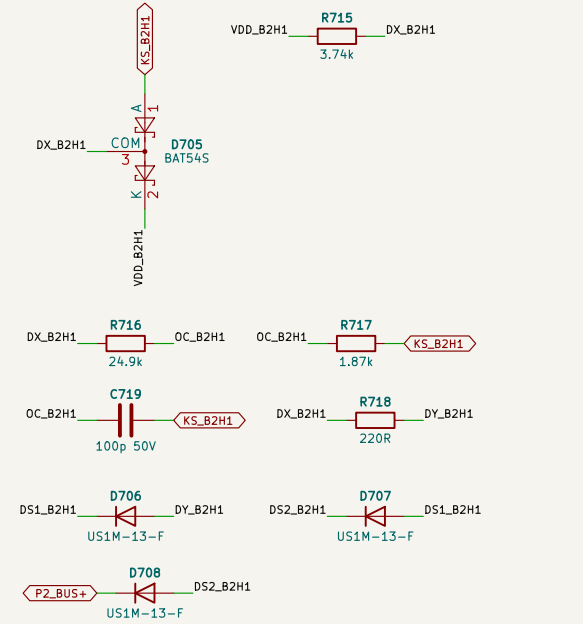
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

B2H1: DRIVER AND GATE



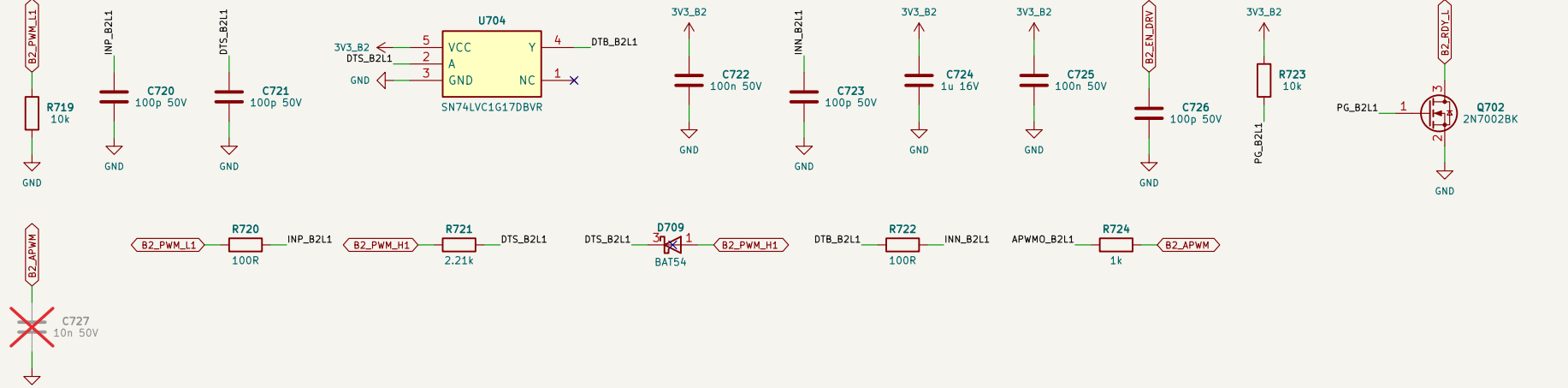
CLMPI direct to the gate (active Miller clamp). Kelvin source = COM.
Zener pair clamps the gate to about +17 V / -6.6 V when the driver cannot.

B2H1: DESAT ON OC



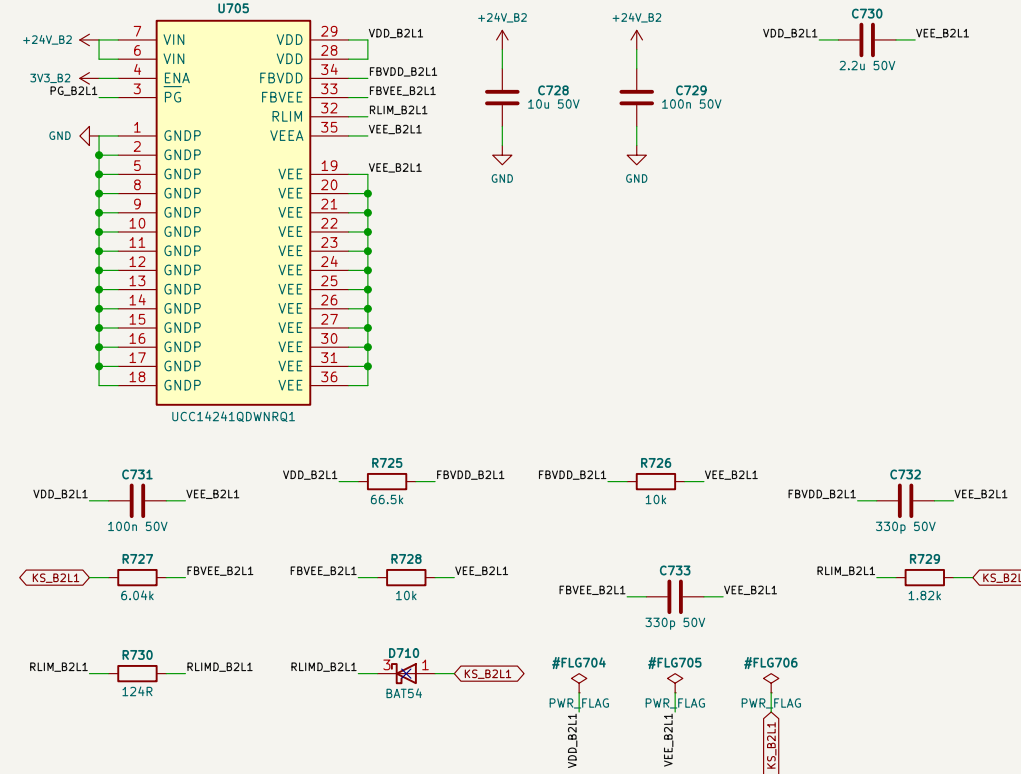
VDD-R1-DX; DX-R2-OC-R3/Cb1k-COM; DX-Rs-3 x US1M-drain (SLUSD43B Fig 9-8). BAT54S keeps DX between COM and VDD against dv/dt through the string.

B2L1: LOGIC SIDE



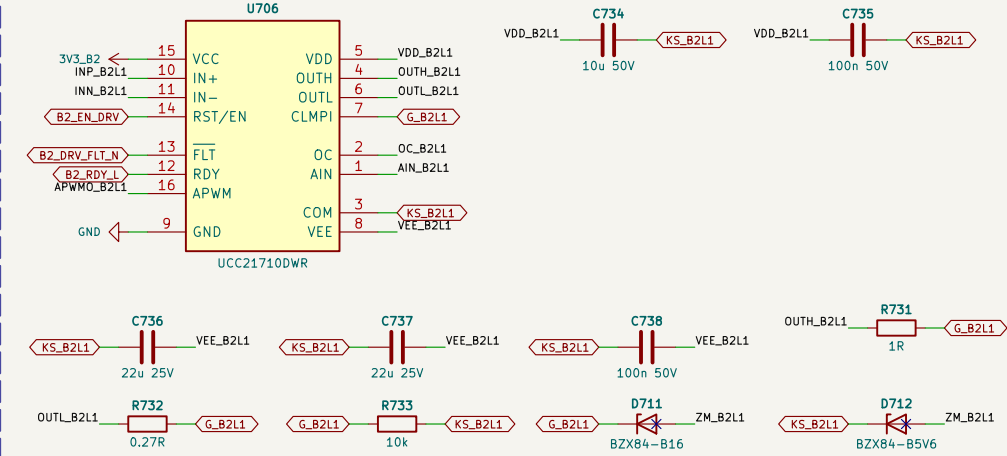
IN+ = own PWM; IN- = complementary switch, falling edge stretched 2.21k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

B2L1: ISOLATED BIAS +15 V / -4 V



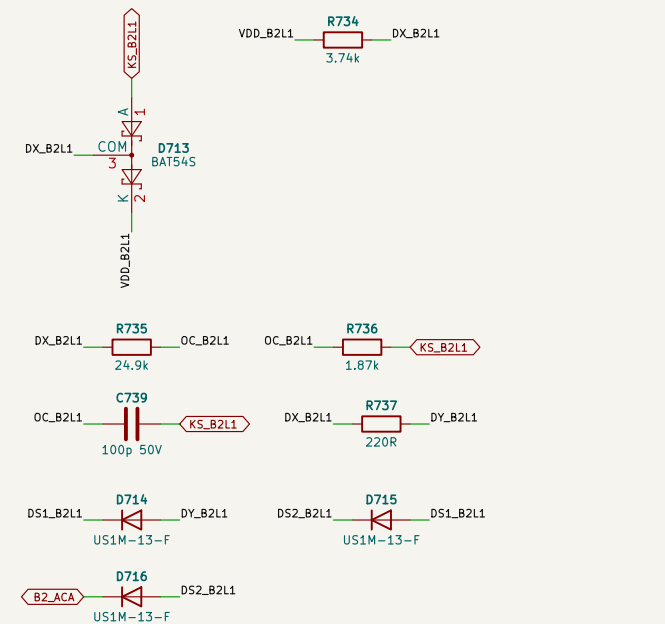
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

B2L1: DRIVER AND GATE



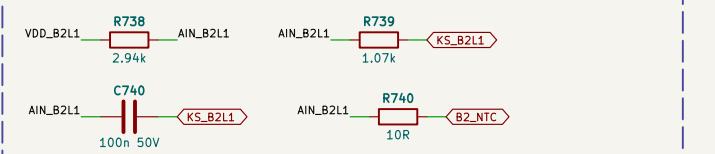
CLMPI direct to the gate (active Miller clamp). Kelvin source = COM.
Zener pair clamps the gate to about +17 V / -6.6 V when the driver cannot.

B2L1: DESAT ON OC



VDD-R1-DX; DX-R2-OC-R3/CbIk-COM; DX-Rs-3 x US1M-drain (SLUSD43B Fig 9-8)
BAT54S keeps DX between COM and VDD against dv/dt through the string.

B2L1: NTC -> AIN



VDD=2.94k-AIN=1.07k-COM biases the NTC ~10x harder than the 203 uA source alone; AIN stays inside 0.6-4.5 V (design_check), -0.5 %/C APWM duty at 100 C.

Not bench-validated. Domains PELV, HV1, HV2, PE, GD_B<sw> (8 gate drivers).
Sheet 8
Schematics authored by Chinmoy Bhuyan
Interfaces: CELL x2 (C1, C2), PORT, SYS-I/O-AUX harness + coolant loop (gen/interfaces.py)

Sheet: /07 Bridge 2 gate drive leg A/
File: 07_b2_gatea.kicad_sch

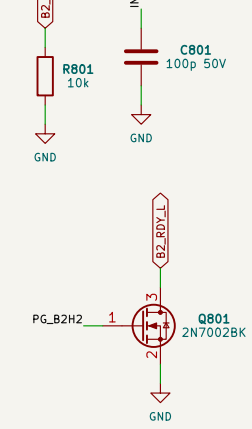
Title: DAB60 power assembly – Bridge 2 gate drive leg A

Size: A2	Date: 2026-10-04
KIC-d.EDA-10.0.6	

Rev: B0
Id: 9.4.7

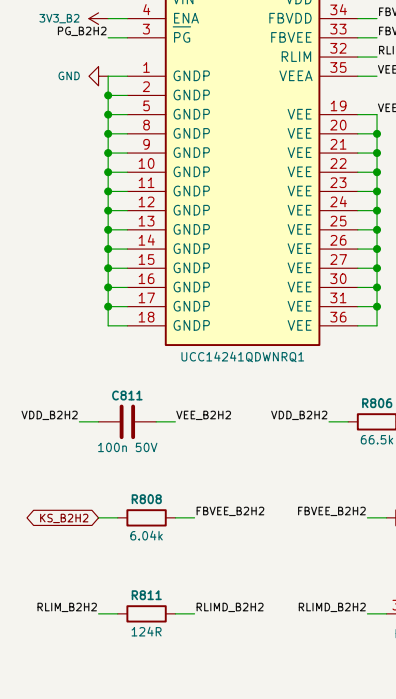
Die VGS of the OFF switch while its partner turns on: -0.55 V at 56.7 V/ns (dab_spec, LM 2.0 A), +0.87 V at 80 V/ns (module max, 2.9 A) vs VGS(th) 1.44 V at 175 C: fine at the DAB rate, marginal at 80 V/ns (gdrv's own case, Roff 1 ohm: +1.19 V). Crss(800 V) understates the low-voltage Cgd: confirm by DPT.

82



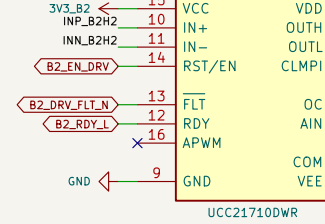
IN+ = own PWM; IN- = complementa
(interlock + min dead time = max(fin

3V3_B2	4	VIN	VDD	34	FBV
PG_B2H2	3	ENA	FBVDD	33	FBV
		PG	FBVEE	32	



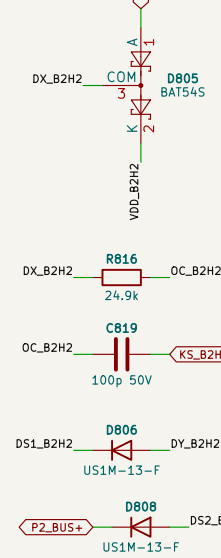
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced by
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

3V3_B2	15	VCC	VDD
INP_B2H2	10	IN+	OUTH
INN_B2H2	11	IN-	OUTI

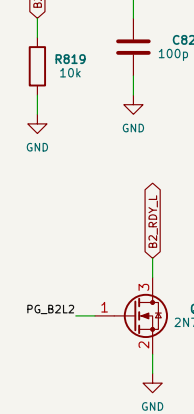


CLMPI direct to the gate (active Miller clamp). K
Zener pair clamps the gate to about +17 V / -

AI

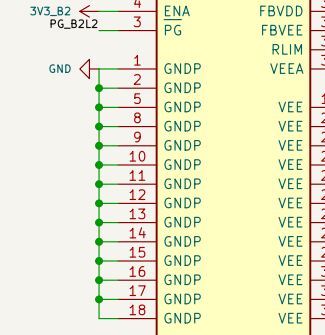


VDD-R1-DX; DX-R2-OC-R3/Cb
BAT54S keeps DX between COM a



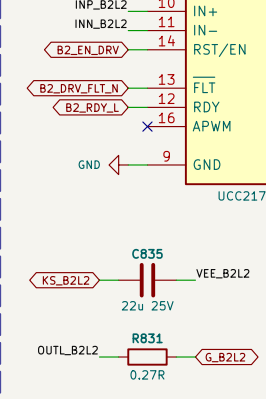
IN+ = own PWM; IN- = com
(interlock + min dead time)

3V3_B2	4	ENA	FBVDD
PG_B2L2	3	PG	FBVEE
			BUM

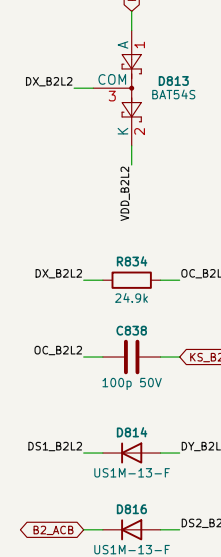


UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, re
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

INP_B2L2	10	IN+
INN_B2L2	11	IN-
OUT_B2L2	14	OUT



CLMPI direct to the gate (active Miller Zener pair clamps the gate to about +



VDD-R1-DX; DX-R2-OC-R3/Cb
BAT54S keeps DX between COM

Not bench-validated. Domains PELV, HV1, HV2, PE, GD_B<sw> (8 gate drivers).
Sheet 9
Schematics authored by Chinmoy Bhuyan
Interfaces: CELL x2 (C1, C2), PORT, SYS-I/O-AUX harness + coolant loop (gen/interfaces.py)

Sheet: /08 Bridge 2 gate drive leg B/
File: 08_b2_gateb.kicad_sch

Title: DAB60 power assembly – Bridge 2 gate drive leg B

Size: A3	Date: 2026-10-04	Rev: B0
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Size: A2	Date: 2020-10-04	Rev: 00
KiCad EDA 10.0.6		Id: 9/17

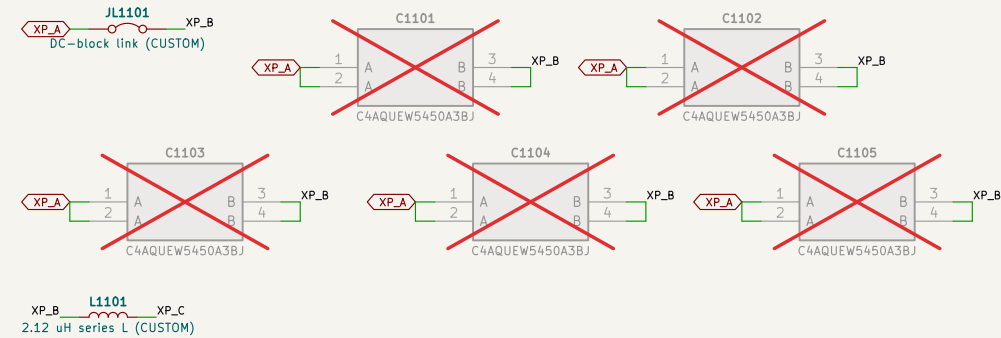
The diagram illustrates the pinout for the 61204021621 component, showing two different pin configurations. The left configuration shows pins 1-40 with various signals, including +24V_B2, GND, B2_PWM1_P, B2_PWM2_P, B2_PWM3_P, B2_PWM4_P, B2_FIL_P, B2_EN_P, B2_RDY, B2_AN1_P, B2_AN2_P, B2_AN3_P, AGND, B2_FIL_N, B2_PWM1_N, B2_PWM2_N, B2_PWM3_N, B2_PWM4_N, B2_ID, B2_AN1_N, and AGND. The right configuration shows pins 1-40 with signals like +24V_B2, GND, PWR_FLAG, +24V_B2, and B2_FIL_N, B2_PWM1_N, B2_PWM2_N, B2_PWM3_N, B2_PWM4_N, B2_ID, B2_AN1_N, and AGND. A resistor R901 (40.2k) is connected between pins 19 and 20.

[illegible][illegible]

LOCAL 5 V (LMR36015) AND 3.3 V (TPS7A2033) FROM THE CELL +24V

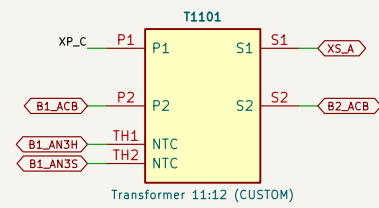
The schematic diagram illustrates the power supply section of the LMR3601SARNXR. The LMR3601SARNXR is shown with its internal pin connections: VIN (pin 2), SW (pin 12), BOOT (pin 7), FB (pin 4), PG (pin 8), VCC (pin 5), PGND (pin 1), AGND (pin 6), and GND (pin 11). The external components include capacitors C913 (4.7u 50V), C914 (220n 50V), C916 (1u 16V), C917 (22u 10V), C918 (22u 10V), C919 (1u 16V), and C920 (2.2u 16V). Inductors L901 (74439346100) and R924 (100k) are also shown. The output is 3V3_B2, which is connected to the TP903 and the output of the LMR3601SARNXR. The diagram also shows the connection to the FB pin of the LMR3601SARNXR, which is connected to the FB pin of the LMR3601SARNXR.

PRIMARY AC PATH (HV1): LEG A – HOB – LINK – L – P1 / P2 – LEG B

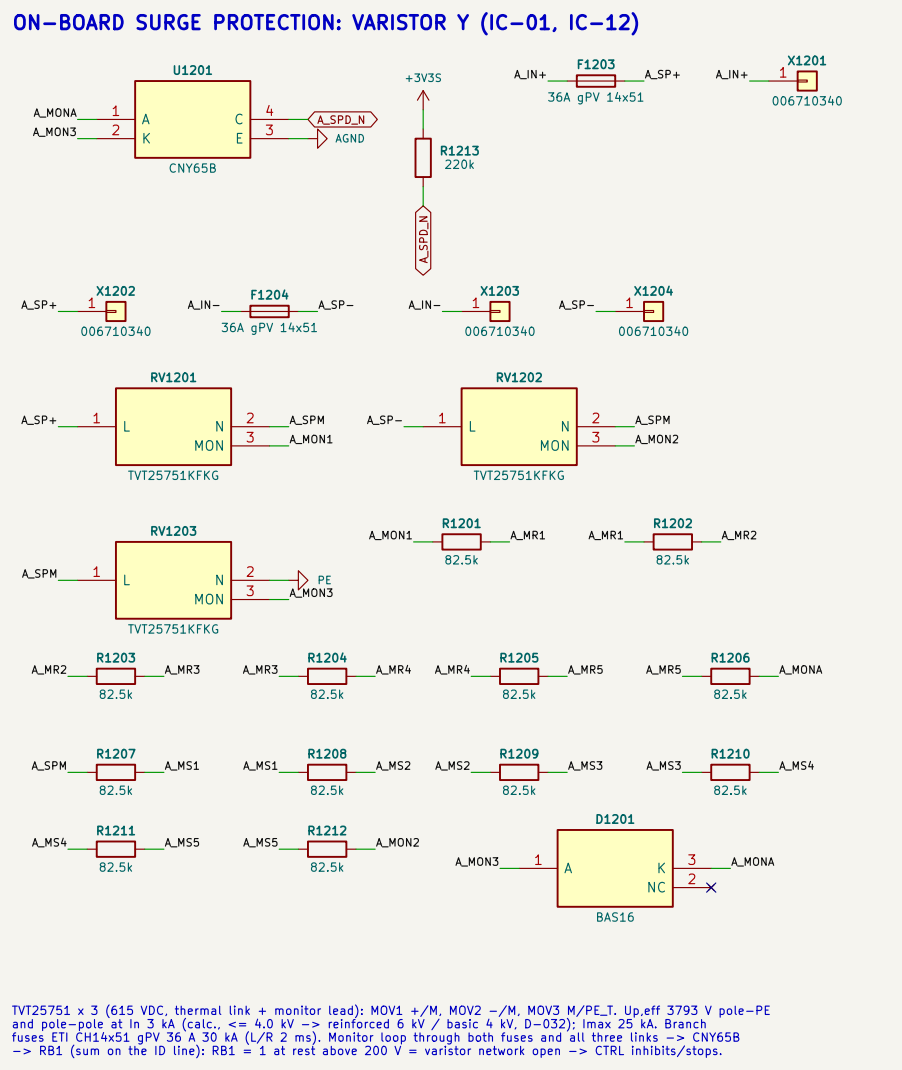
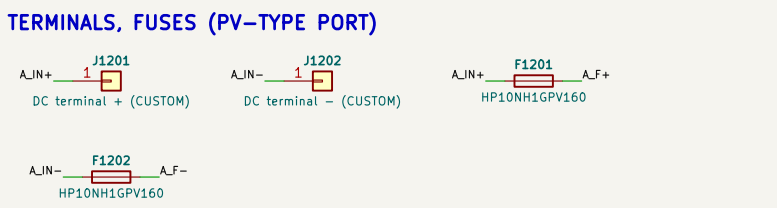


Link FITTED. Provision: unbolt it and fit the blocking bank 5 x C4AQUEW5450A3BJ (225 uF) (DNP below, 225 uF).
Firmware flux-balance loop on AN1 + gapped core (L_m 150 uH) + hardware |L_{dc}| trip is the baseline.
Series L (CUSTOM): 2.12 uH nominal, trimmed 1.24–2.99 uH so that leakage + L = 6.50 uH +/-5 % (leakage + external, measured as a matched set);
119 A rms, 177 A peak, no saturation to 275 A, <= 11 W.

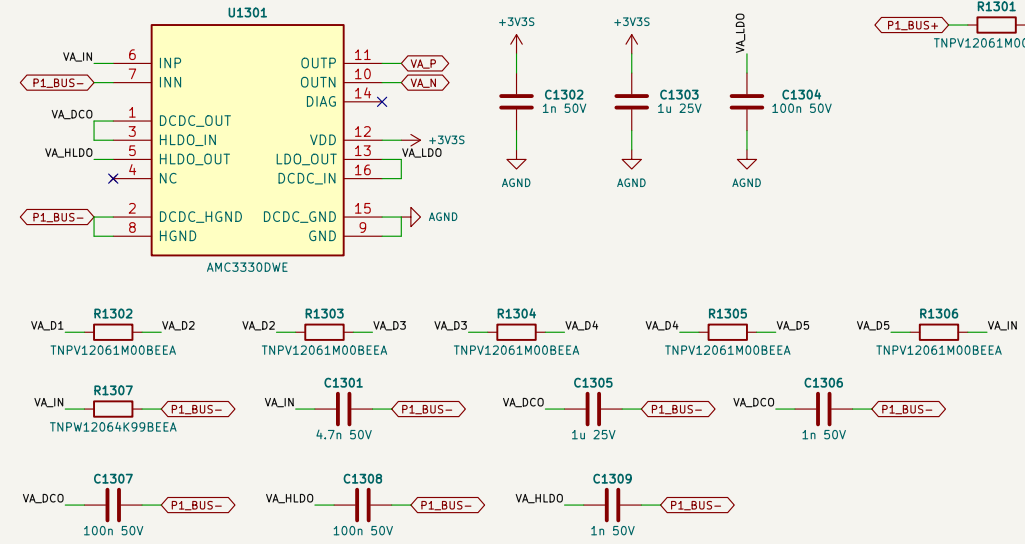
TRANSFORMER 11:12 (HV1 / HV2 BARRIER, CUSTOM)



P1/S1 = dotted ends. Secondary: leg A of bridge 2 – HOB – S1 / S2 – leg B of bridge 2. TH1/TH2: embedded
NTC -> bridge 1 AN3. L_{sigma} 4.38 uH +/-20 %, L_m 150 uH +/-10 %, 119 / 112 A rms, 177 / 162 A peak, <= 159 W,
hot spot <= 130 C. Insulation basic (pending system decision) on a 1850 V DC working basis: hipot/PD from ECO-10, PD <= 10 pC.
Full text in the part description and report.md sec. 9.

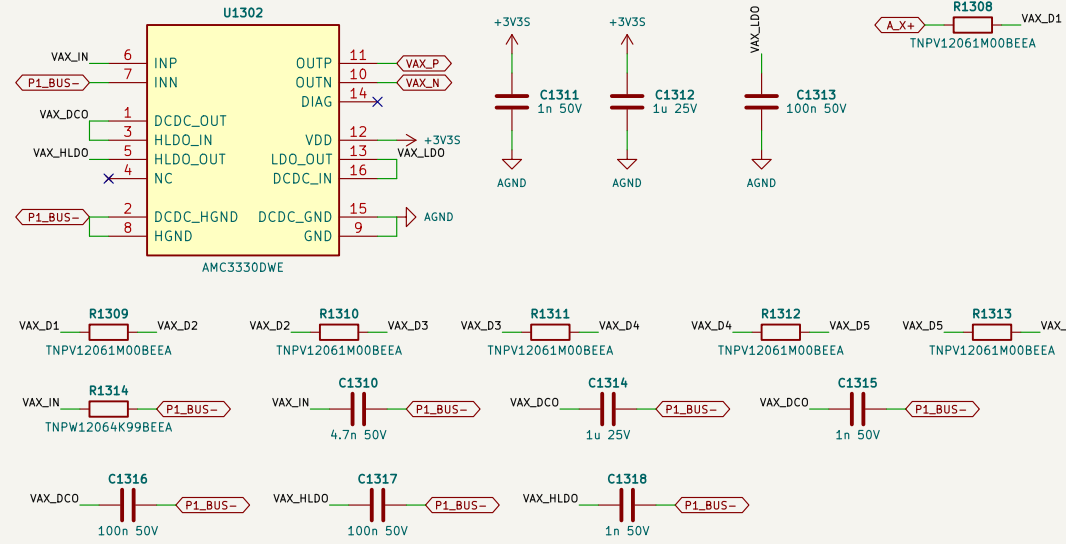


VA – BUS-SIDE VOLTAGE



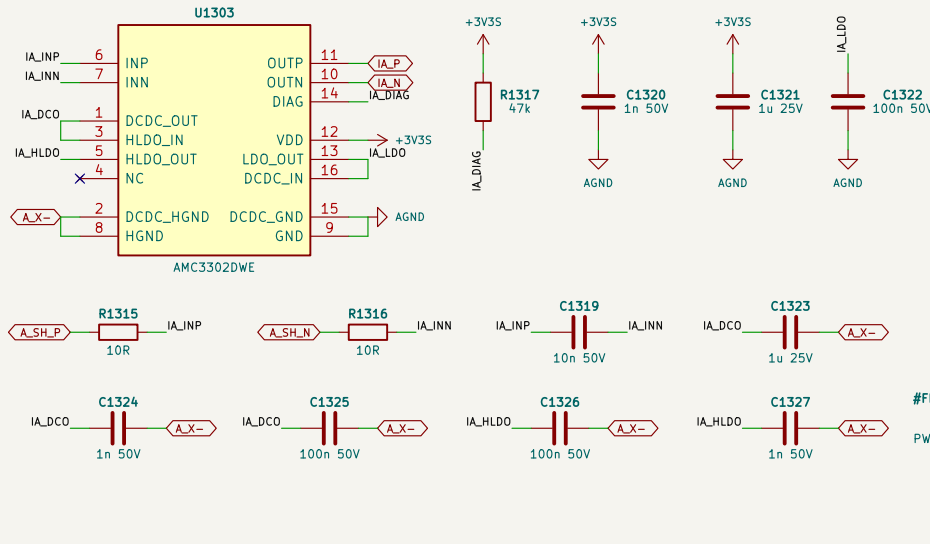
6 × TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k ± 1 V = 1203 V, clip 150k V, 4.7 nF, lag
≤ 27 us incl. AMC3330; Out: VALP/N (±/−2 V on 1.4k V, PORT contract); HGND = P1_BUS−

VAX – TERMINAL-SIDE VOLTAGE



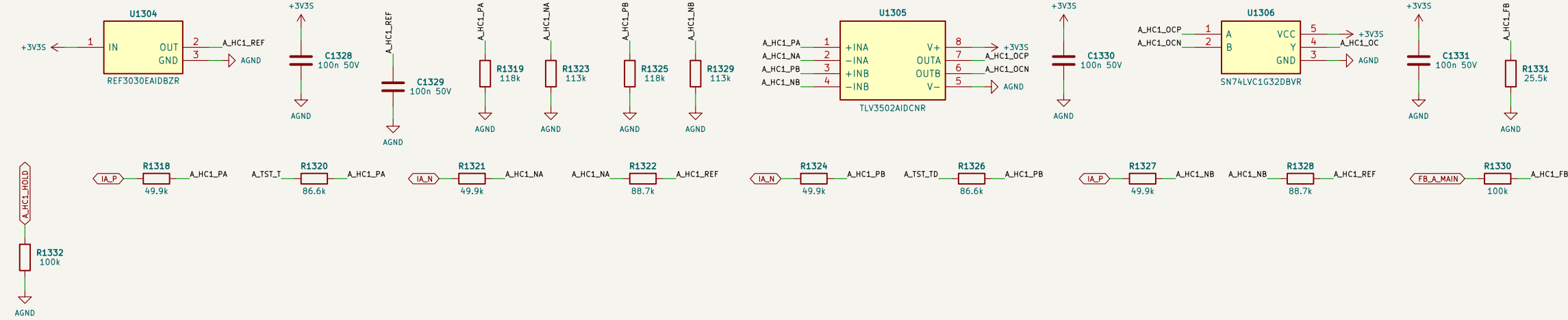
6 × TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k ± 1 V = 1203 V, clip 150k V, 4.7 nF, lag
≤ 27 us incl. AMC3330; Out: VAXP/N (±/−2 V on 1.4k V, PORT contract); HGND = P1_BUS−

IA – PORT CURRENT



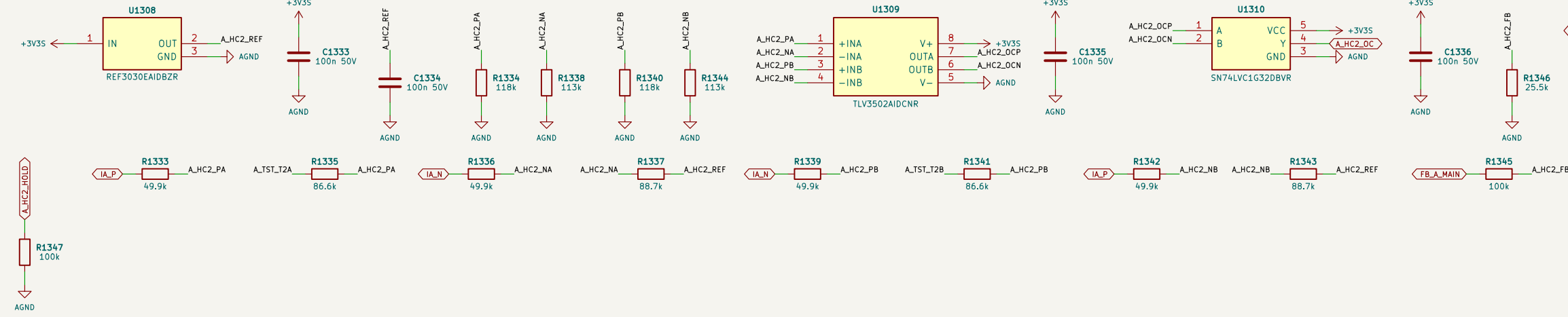
Linear ±/−500 A, clip ±/−640 A (OC trip 202 A, SC trip 405 A inside), 10 ohm + 10 nF input
filter (S8AS4118 Fig 8-2); HGND on the shunt's V− bolt by its own trace. Fail-safe out = −2.57 V.

HOLD-CLOSED LOGIC, CHANNEL 1 (IRFR9214 HIGH SIDE)



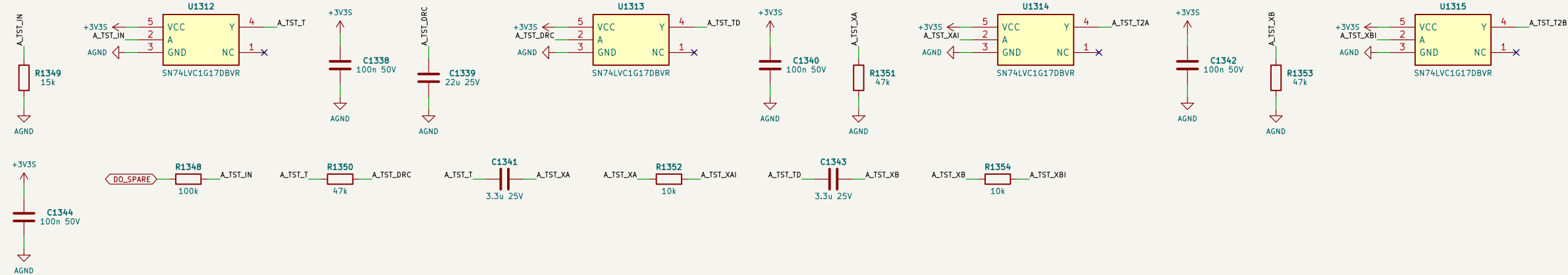
[IA] > 410 A (band 382–441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND
DIAG = valid → HOLD1. Test injection 86.6 k from A_TST_T (side A) / A_TST_TD (side B); shifts the threshold
by +59 A, so a healthy side trips at 0 A under TEST only.

HOLD-CLOSED LOGIC, CHANNEL 2 (IRFL214 LOW SIDE)



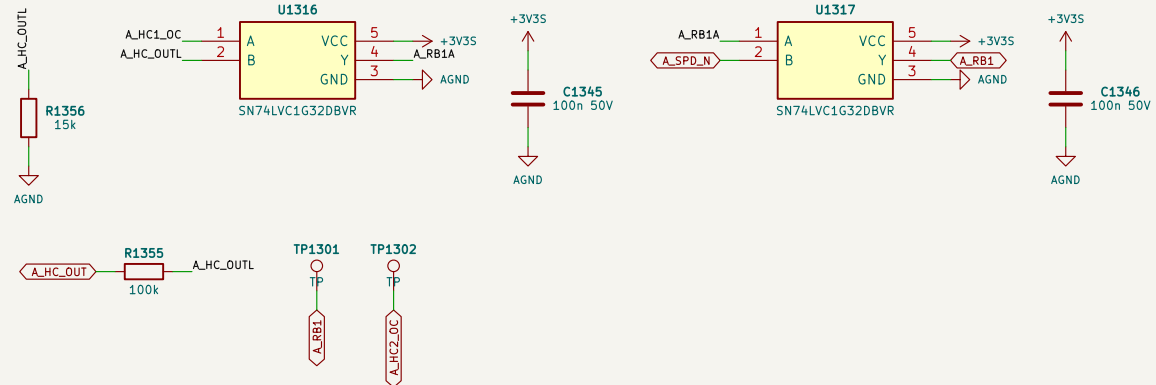
[IA] > 410 A (band 382–441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND
DIAG = valid → HOLD2. Test injection 86.6 k from A_TST_T2A (side A) / A_TST_T2B (side B); shifts the threshold
by +59 A, so a healthy side trips at 0 A under TEST only.

PROOF-TEST INJECTION (DO_SPARE = TEST)



100 k / 15 k → Schmitt buffer T (ch1 side A), T → 47 k / 22 uF → T_d (ch1 side B), One-shots
(2.3 uF / 47 k) at T → T2A (ch2 side A) and of T_d → T2B (ch2 side B). No single fault trips both
channels for longer than one-shot. Firmware runs it at every start, contactor open and closed
(report sec. 7.4: windows, what it finds and what it does not).

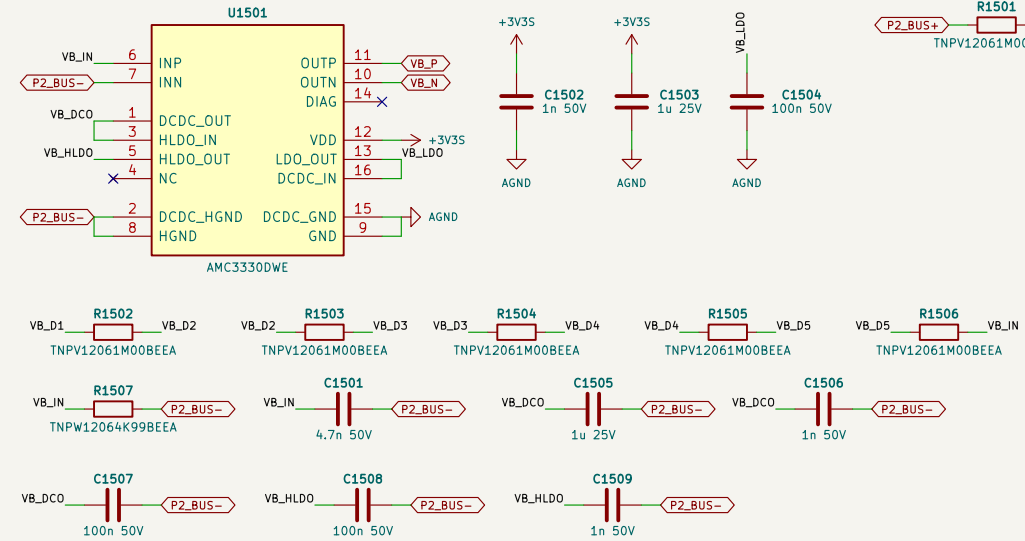
READBACK RB1 / RB2 (TO CTRL ON A SPARE LINE)



RB1 = OC1 OR hold high-side output present (100 k / 15 k); at rest RB1 = 1 means a stuck channel 1.
RB2 = OC2, PV=PORT sums them onto the ID line (boardLo); Firmware reads RB1 + RB2 against the
measured current in operation and through the proof-test windows at start.

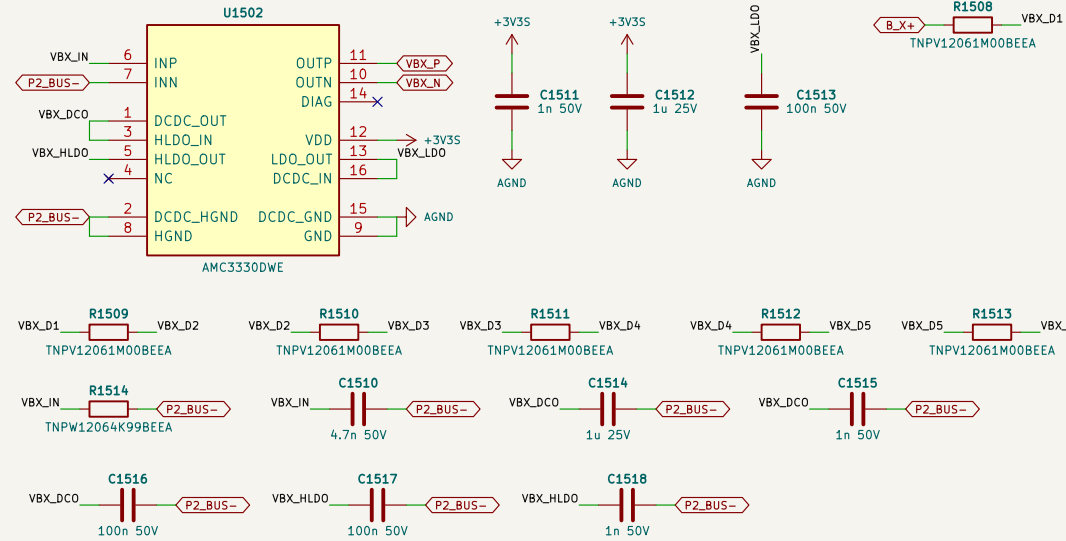
Not bench-validated. Domains PELV, HV1, HV2, PE, GD_B<sw> (8 gate drivers),
Sheet 14
Schematics authored by Chimoy Bhuyan
Interfaces: CELL x2 (C1, C2), PORT, SYS=IO=AUX harness + coolant loop (gen/interfaces.p)

VB – BUS-SIDE VOLTAGE



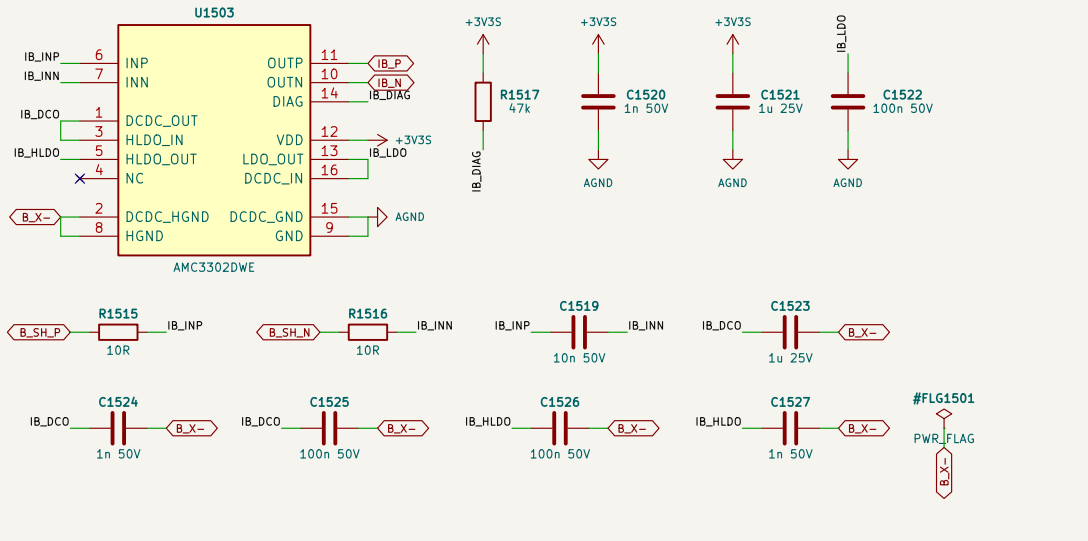
6 × TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF, lag
≤ 27 us incl. AMC3330; Out: VB_L/P/N (±/−2 V on 1.44 V, PORT contract); HGND = P2_BUS−

VBX – TERMINAL-SIDE VOLTAGE



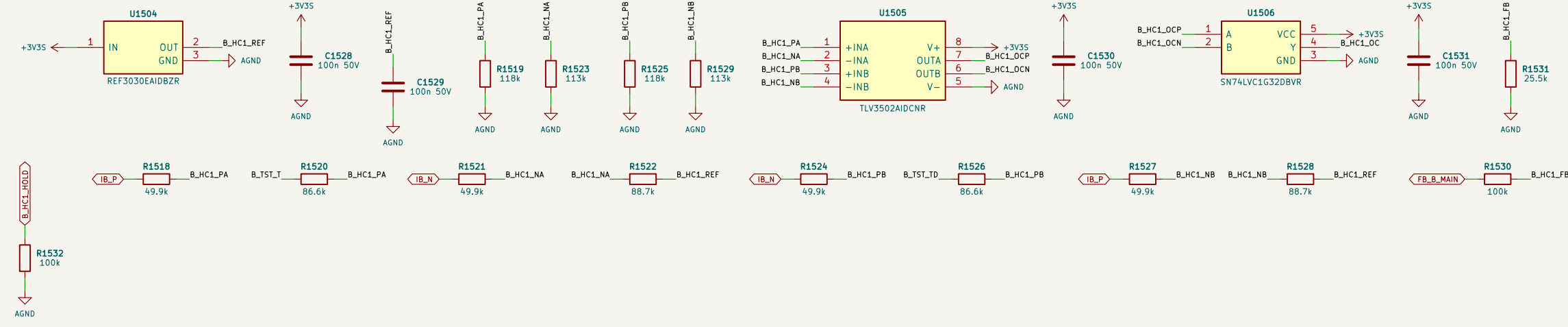
6 × TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF, lag
≤ 27 us incl. AMC3330; Out: VBX_L/P/N (±/−2 V on 1.44 V, PORT contract); HGND = P2_BUS−

IB – PORT CURRENT



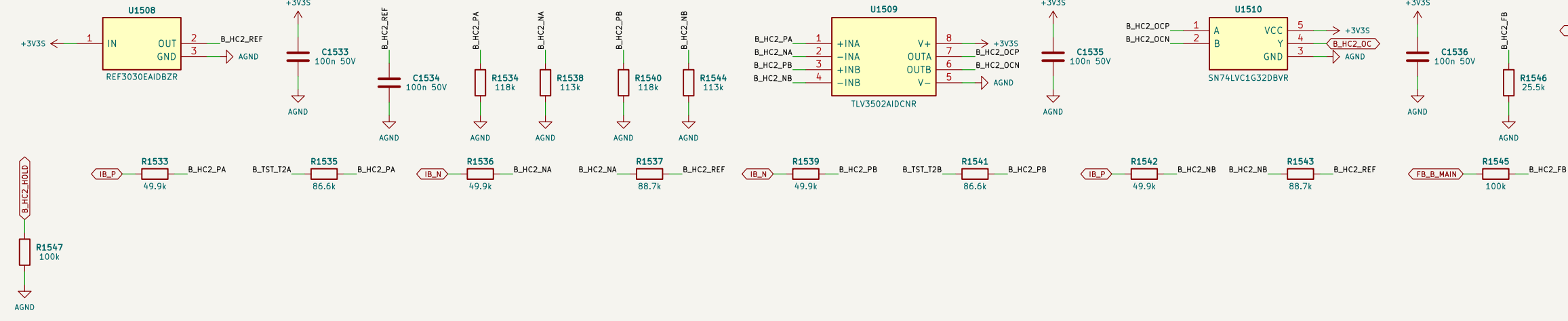
Linear ±/−500 A, clip ±/−640 A (OC trip 202 A, SC trip 405 A inside), 10 ohm + 10 nF input
filter (S8A5A11B Fig 8-2); HGND on the shunt's K− bolt by its own trace. Fail-safe out = −2.57 V.

HOLD-CLOSED LOGIC, CHANNEL 1 (IRFR9214 HIGH SIDE)



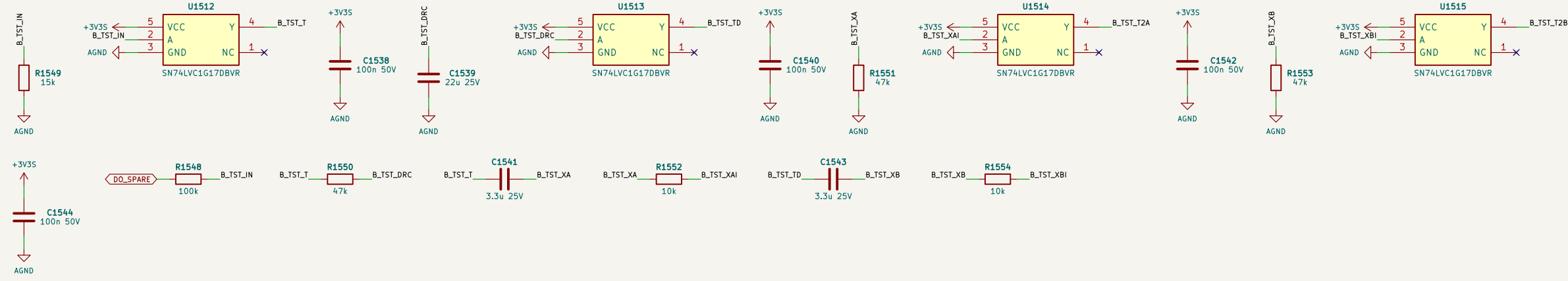
|IB| > 410 A (band 382–441 A, own REF3030E + 0.1 % network, CM–cancelling) AND FB = closed AND
DIAG = valid → HOLD1. Test injection 86.6 k from B_TST_T (side A) / B_TST_TD (side B); shifts the threshold
by +59 A, so a healthy side trips at 0 A under TEST only.

HOLD-CLOSED LOGIC, CHANNEL 2 (IRFL214 LOW SIDE)



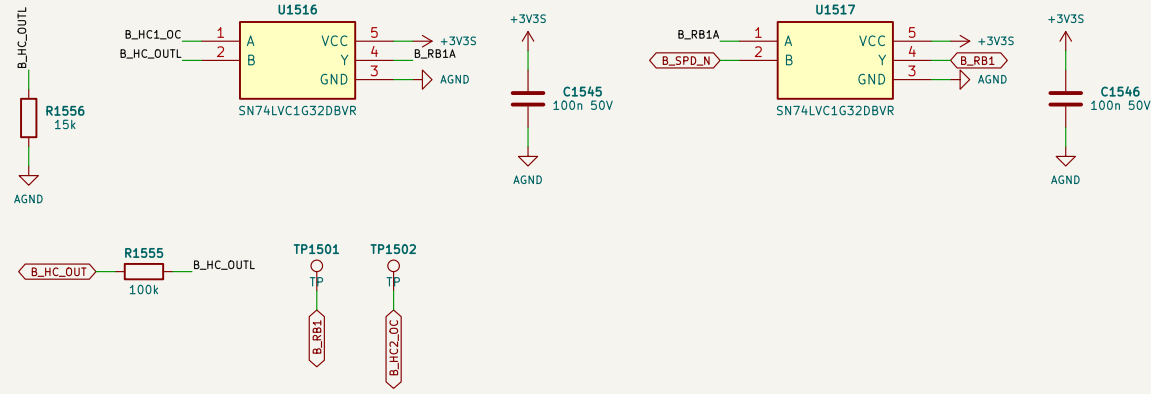
|IB| > 410 A (band 382–441 A, own REF3030E + 0.1 % network, CM–cancelling) AND FB = closed AND
DIAG = valid → HOLD2. Test injection 86.6 k from B_TST_T2A (side A) / B_TST_T2B (side B); shifts the threshold
by +59 A, so a healthy side trips at 0 A under TEST only.

PROOF-TEST INJECTION (DO_SPARE = TEST)



100 k / 15 k → Schmitt buffer T (ch1 side A); T → 47 k / 22 uF → T_d (ch1 side B). One–shots
(2.3 uF / 47 k) at T → T2A (ch2 side A) and of T_d → T2B (ch2 side B). No single fault trips both
channels for longer than one–shot. Firmware runs it at every start, contactor open and closed
(report sec. 7.4: windows, what it finds and what it does not).

READBACK RB1 / RB2 (TO CTRL ON A SPARE LINE)



RB1 = OC1 OR hold high–side output present (100 k / 15 k); at rest RB1 = 1 means a stuck channel 1.
RB2 = OC2; PV–PORT sums them onto the ID line (boardIo); Firmware reads RB1 + RB2 against the
measured current in operation and through the proof–test windows at start.

Not bench–validated. Domains PELV, HV1, HV2, PE, GD_B<sw> (8 gate drivers).
Sheet 16
Schematics authored by Chimoy Bhuyan
Interfaces: CELL x2 (C1, C2), PORT, SYS–IO–AUX harness + coolant loop (gen/interfaces.p)

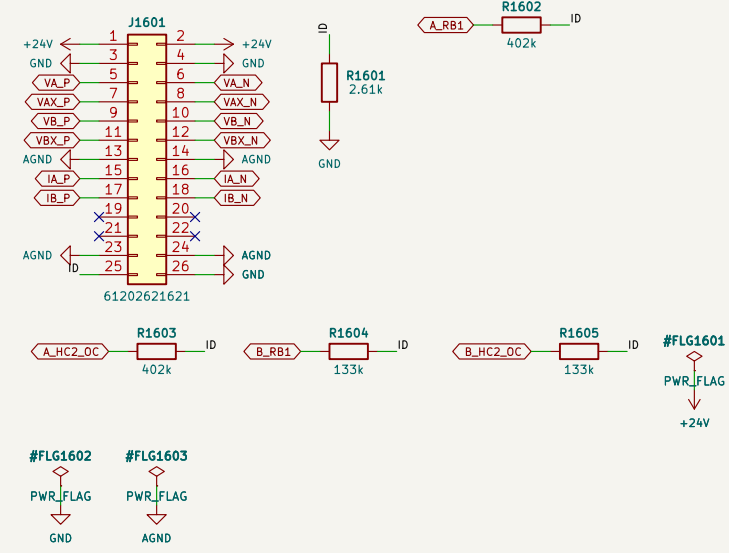
Sheet: /15 Port B sensing and hold logic/
File: 15_portB_sense.kicad_sch

Title: DAB60 power assembly – Port B sensing and hold logic

Size: A2 Date: 2026–10–04 Rev: B0

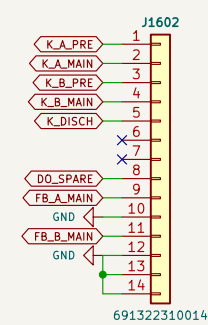
KiCad E.D.A. 10.0.6 Id: 16/17

PORT CONNECTOR (CTRL-C2000), ID WITH HOLD READBACK



interfaces.PORT: VA/VAX/IA = port 1 (tag A), VB/VBX/IB = port 2 (tag B). AUX1/AUX2 not used (no IMD).
ID: 2.61k || readback = 2.480 k (code 2.49k). RB1/RB2 of port A via #02k, of port B via 133k: nine levels
0.496–0.632 V within ± 165 mV of the code (PVCELL–25 window starts at 0.667 V); unpowered 0.517 V.

SYS-IO-AUX HARNESS (COILS, CONTACTOR FEEDBACK, PROOF TEST)

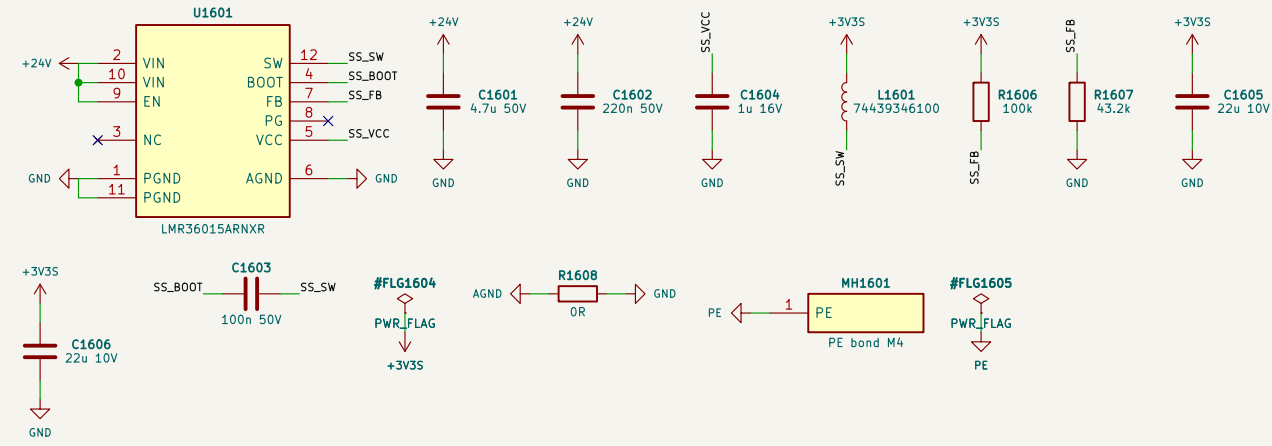


port.CTRL_PINS, 14-way, the PV-PORT harness: DO_SPARE = hold-logic proof test of both ports.
IMD_SW_P/N not used on DAB60 (no insulation measurement): open.

AUX-HV FEED: PORT 1 ONLY (D-021, INT-11)

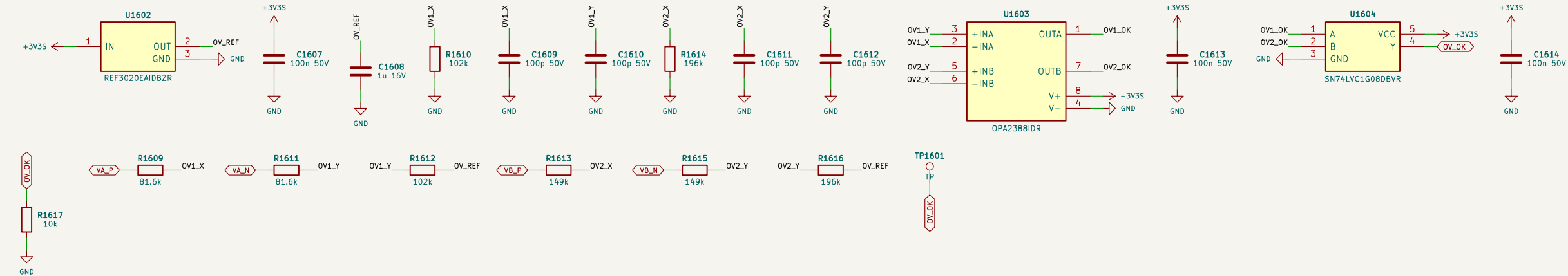
AUX-HV PA+ = J1206 on A_X+ (terminal side of port 1: fuse + EMI filter, ahead of the contactor).
PA- = J1205 on P1_BUS-. AUX-HV PB+/PB- stay open. Port 2 stud J1406 on B_X+ is DNP: no port-2 tap exists, so HV1 and HV2 never meet in the AUX-HV cable OR. Consequence: no black start from port 2 without cabinet 24 V.

SENSOR_SUPPLY +3V3S (PORT_SPEC.JSON SENSOR_SUPPLY)



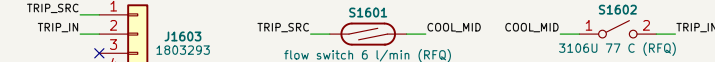
LMR36015ARNXR 400 kHz, 10 uH, 100k / 43.2k -> 3.315 V for 4 x AMC3330 + 2 x AMC3302 + port hold logic + OV comparator (SNV5849D Fig 10-1: 4.7 uF + 220 nF in, 100 nF boot, 1 uF VCC, 2 x 22 uF out). AGND joins GND only here.

PORT OVER-VOLTAGE COMPARATORS (BOTH BRIDGES TRIP, LATCHED THERE)



Matched dividers on the AMC3330 outputs: OK while $V_{AP} - V_{AN} < 2.048 \text{ V} \times R1/R2$ (common mode rejected); OPA2388 as comparator (VOS 7.5 uV; inputs differ by mV at the trip point). Port 1 trips at 986 V (972–1000 V), port 2 at 937 V (924–950 V); worst case $\approx$ the spec $1000 \text{ V} / 950 \text{ V}$, above the firmware 960 V / 920 V. Response 37.5 us (divider 4.9k Ω), 4.7 nF + AMC3330 27k uV; OPA2388 overload recovery 10 us, logic; load-rejection peak 1029 V / 970 V, OV_OK low = 1; trip pull-down: not +2V35 (PORT cable out) trips both.

COOLANT LOOP -> SYS-IO-AUX J801 SPARE TRIP INPUT (ISO1211, 1:1 HARNESS)



Flow switch (closed while flow > 6 l/min) in series with the cold-plate thermostat (opens at 77 C):
 loop open = channel-A hardware trip on SYS-IO-AUX (latched there, no firmware). SYS wets the loop from 24V_OR
 via 2.2k into the ISO1211 limiter: 0.00-0.00 mA, <= 30 V open. Contacts: reed (VK3, 1 A / 48 V) and gold
 WE-1 (3106U, 500 mA / 50 V, low-level rated). Pins 3/4 (SYS GND) unused here.